Computer Networking



谢 逸
中山大学 • 计算机学院

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Chapter 1 Introduction

A note on the use of these PowerPoint slides:

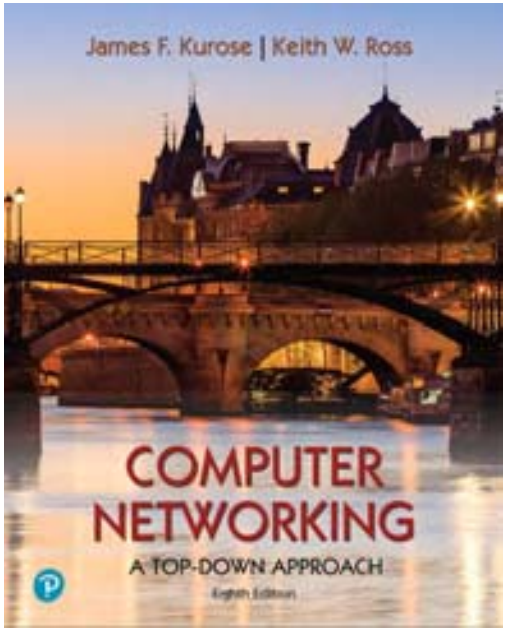
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For a revision history, see the slide note for this page.

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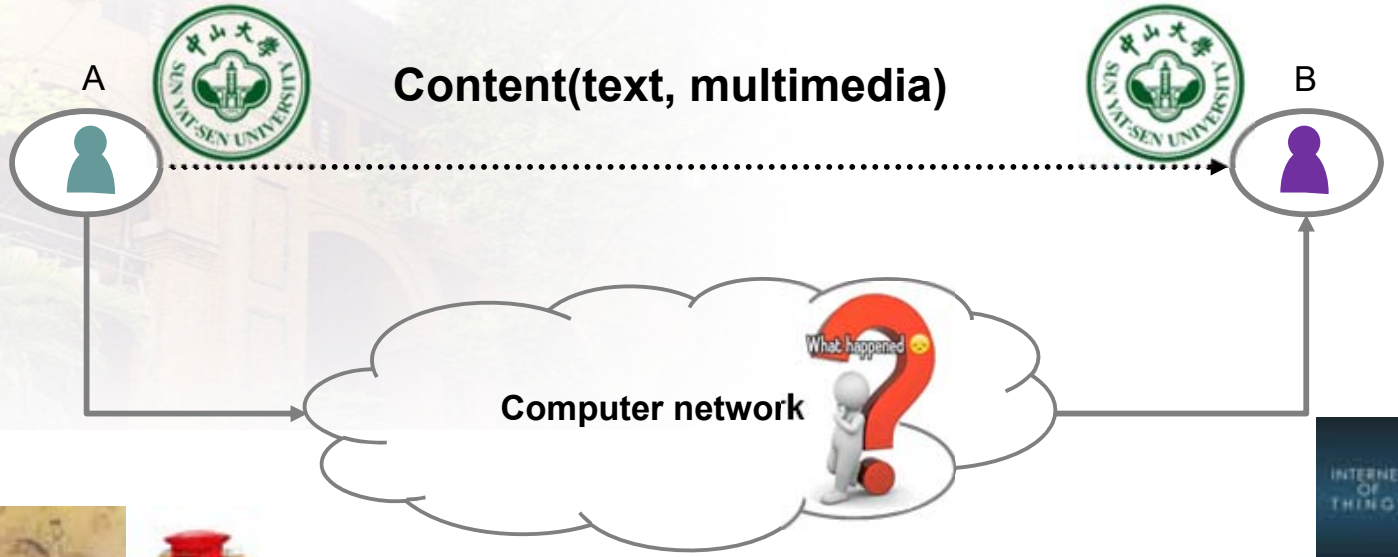
Computer Networking: A Top-Down Approach
8th edition
Jim Kurose, Keith Ross
Pearson, 2020

Computer Network is different from other fields:

- Computer networks are artificial systems, not natural products, , which leads to the existence of a large number of artificial concepts；
- The network is large, complex, and time-varying, making it difficult to describe using general mathematical models；
- It involves low-level physical networks, middle-level data analysis, and high-level behavior understanding；
- Data-driven analysis is only a part of it, while physical mechanisms and principles are more important。
- Only by mastering the underlying core technology can one gain the right to speak。

The first question:

What happens to the network in end-to-end communication?







How to achieve it?





















Assignments:

- Ch1(ver7, CN/EN): 6, 7, 8, 9, 10, 13, 14, 20, 25, 27, 29, 31, 33
- Ch1(ver8, CN/EN): 6, 7, 8, 9, 10, 13, 14, 20, 25, 27, 29, 31, 33
- Practice: 18, 19

下横杠题目为必做题

- Keywords : protocol, network edge, access network, Packet Switching, Circuit Switching, delay, Encapsulation

Chapter 1: introduction

1.1 what *is* the Internet?

1.2 network edge

- end systems, access networks, links

1.3 network core

- packet switching, circuit switching, network structure

1.4 delay, loss, throughput in networks

1.5 protocol layers, service models

1.6 networks under attack: security

1.7 history

The Internet: a “nuts and bolts” view



Billions of connected computing *devices*:

- *hosts* = *end systems*
- running *network apps* at Internet’s “edge”



Packet switches: forward packets (chunks of data)

- *routers*, *switches*

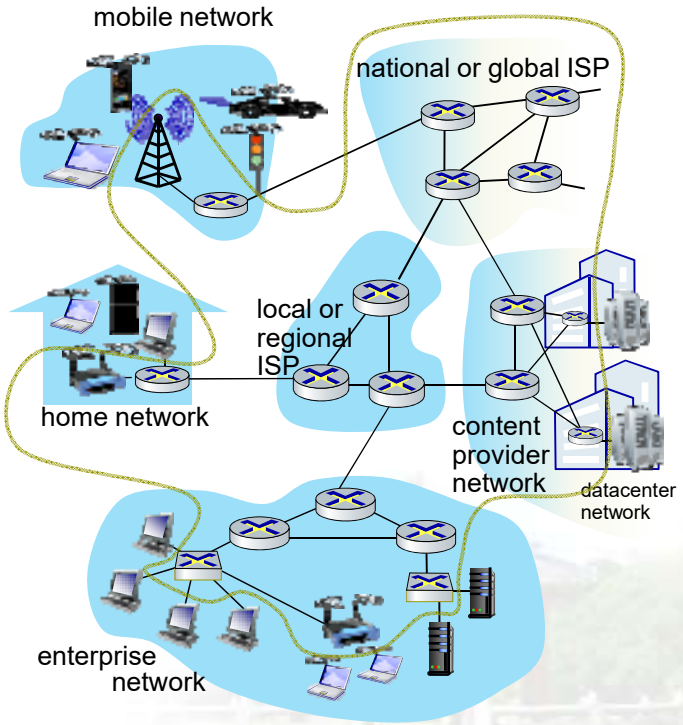


Communication links

- fiber, copper, radio, satellite
- transmission rate: *bandwidth*

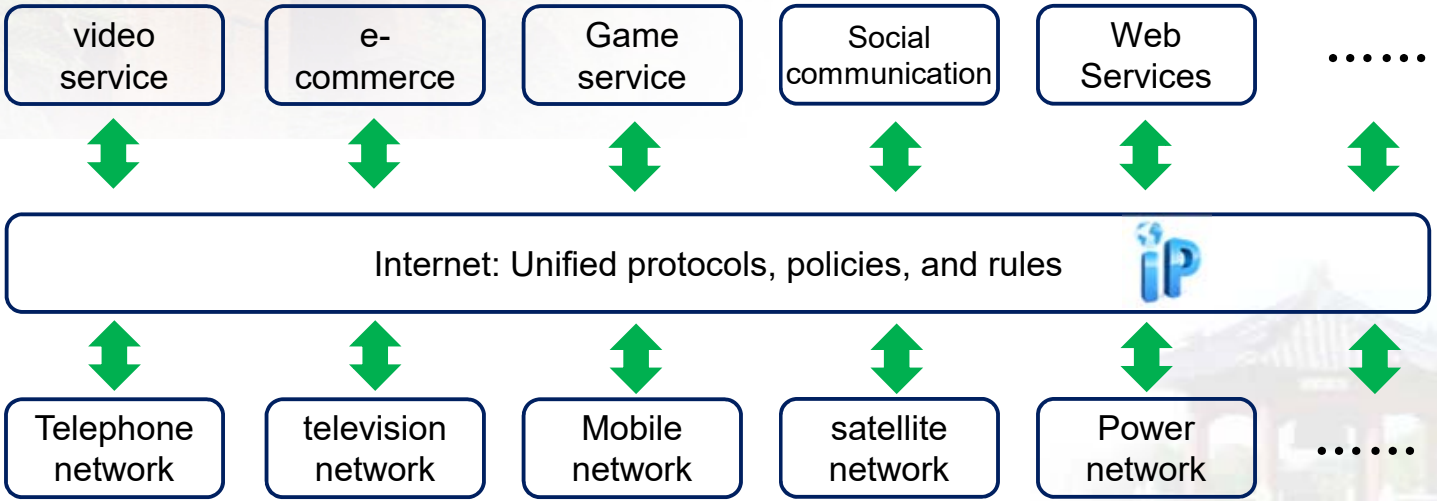
Networks

- collection of devices, routers, links: managed by an organization



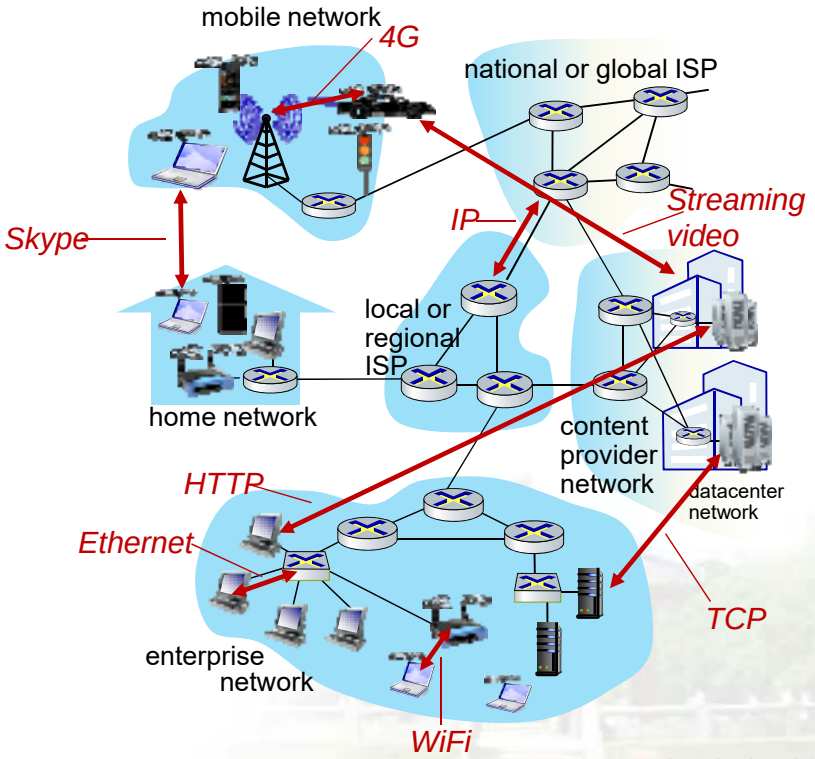
The Internet: a “nuts and bolts” view

- *Internet:* “network of networks”
  - Interconnected ISPs



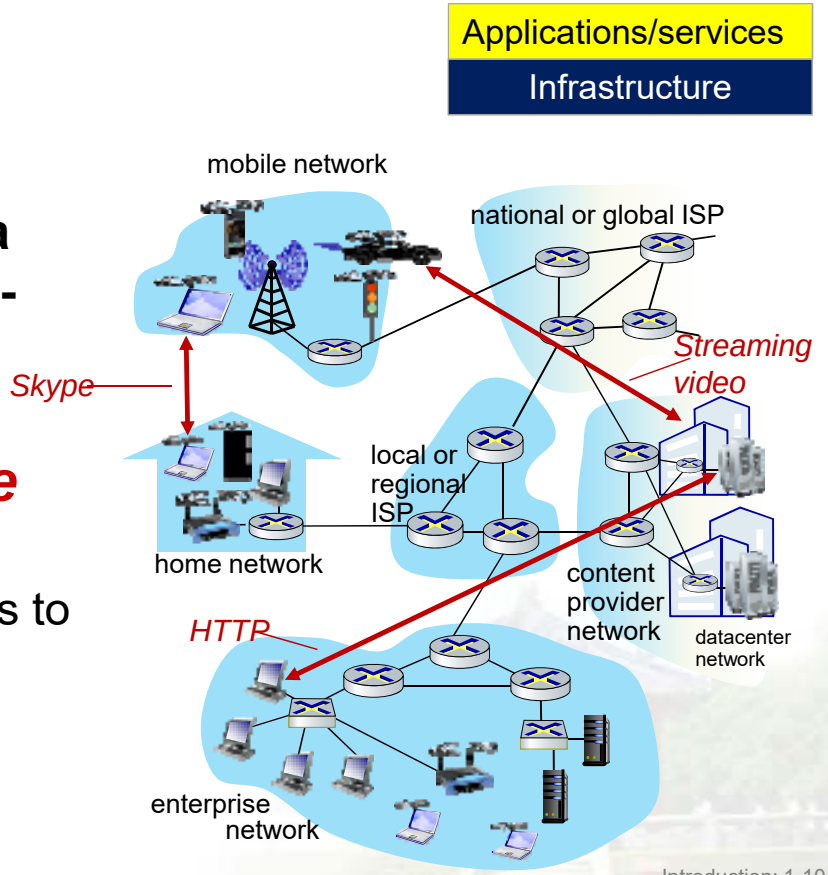
The Internet: a “nuts and bolts” view

- ***protocols are everywhere***
  - control sending, receiving of messages
  - e.g., HTTP (Web), streaming video, Skype, TCP, IP, WiFi, 4G, Ethernet
- ***Internet standards***
  - RFC: Request for Comments
  - IETF: Internet Engineering Task Force



The Internet: a "services" view

- ***Infrastructure*** **that provides services to applications:**
  - **Web, streaming video, multimedia teleconferencing, email, games, e-commerce, social media, inter-connected appliances, …**
- ***provides programming interface to distributed applications:***
  - "hooks" allowing sending/receiving apps to "connect" to, use Internet transport service
  - provides service options, analogous to postal service



What's a protocol?

human protocols:

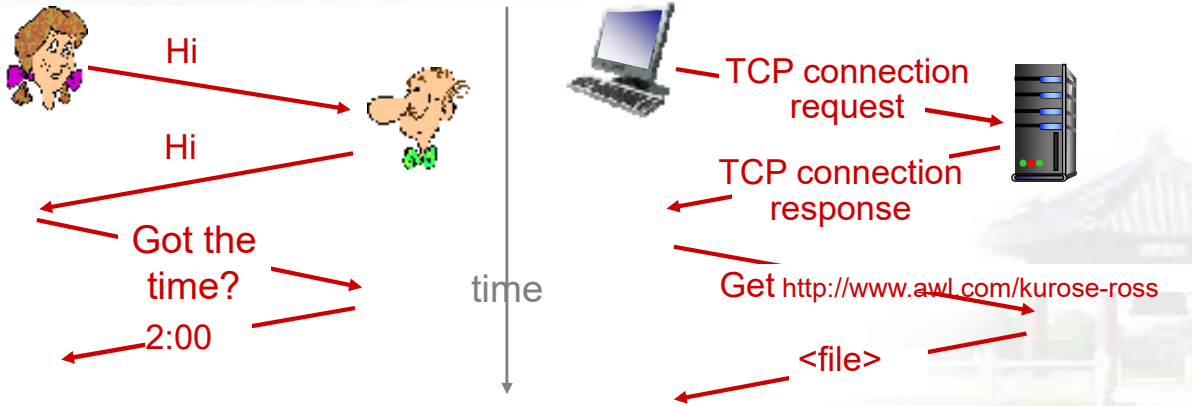
- "what's the time?"
- "I have a question"
- introductions

specific **msgs** sent
specific **actions** taken when msgs received, or other events

network protocols:

- machines rather than humans
- all communication activity in Internet **governed by protocols**

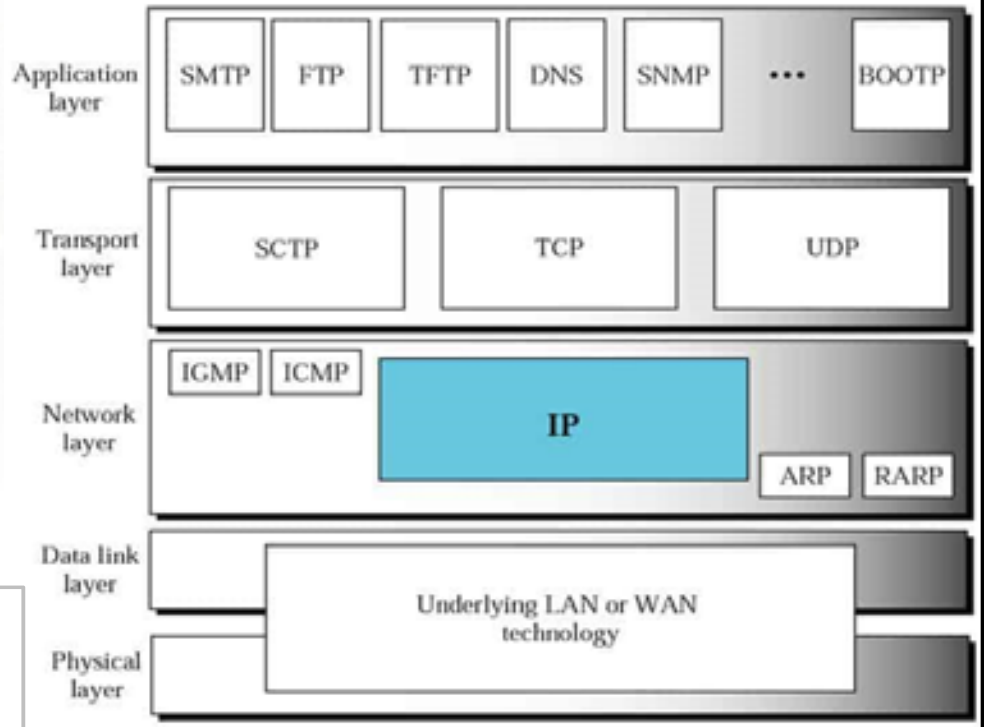
a human protocol **and** **a computer network protocol:**



What's a protocol?

network protocols:

protocols define format, order of msgs sent and received among network entities, and actions taken on msg transmission, receipt



例如: 中国广东省广州市大学城中山大学计算机学院

School of Computer Science and Engineering, Sun Yat-sen University, Guangzhou Higher Education Mega Center, China

计算机大学城广州市中国学院中山大学广东省

Chapter 1: introduction

1.1 what *is* the Internet?

1.2 network edge

- end systems, access networks, links

1.3 network core

- packet switching, circuit switching, network structure

1.4 delay, loss, throughput in networks

1.5 protocol layers, service models

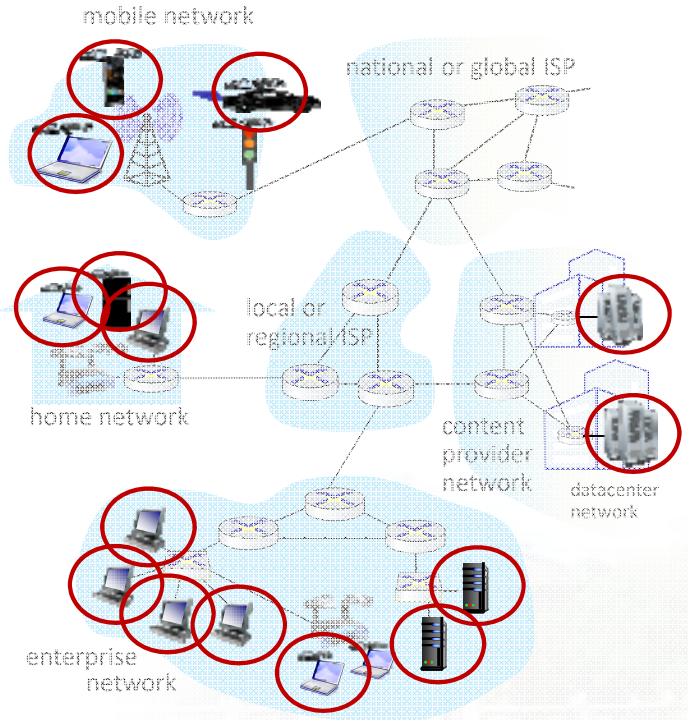
1.6 networks under attack: security

1.7 history

A closer look at Internet structure

Network edge:

- **hosts: clients and servers**
- **servers often in data centers**



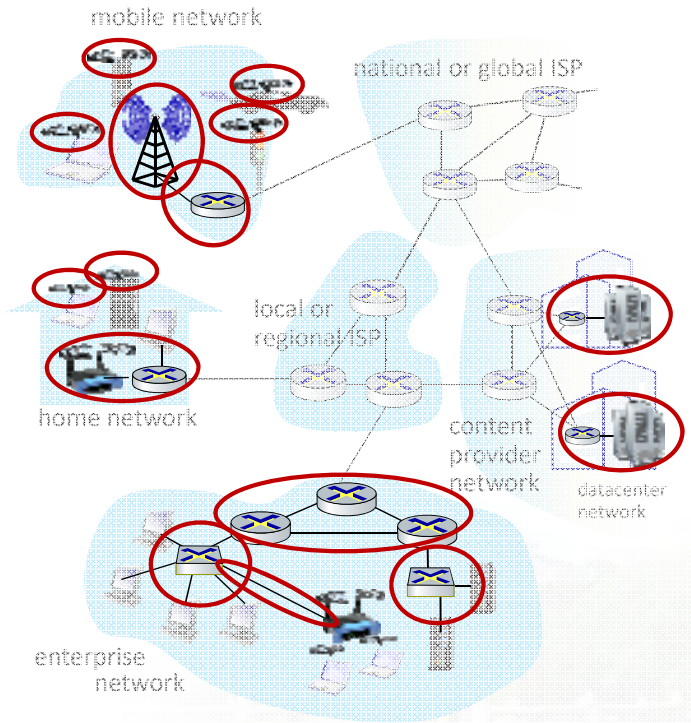
A closer look at Internet structure

Network edge:

- hosts: clients and servers
- servers often in data centers

Access networks, physical media:

- **wired, wireless communication links**



A closer look at Internet structure

Network edge:

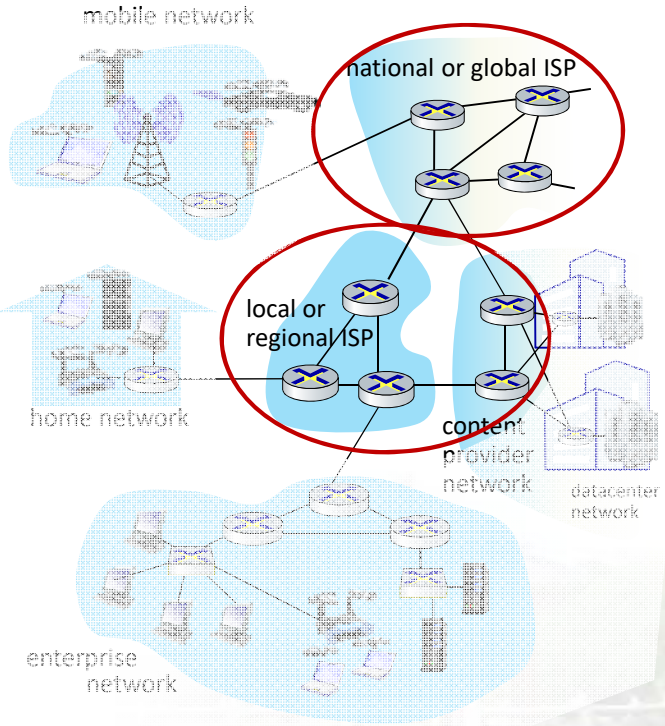
- hosts: clients and servers
- servers often in data centers

Access networks, physical media:

- wired, wireless communication links

Network core:

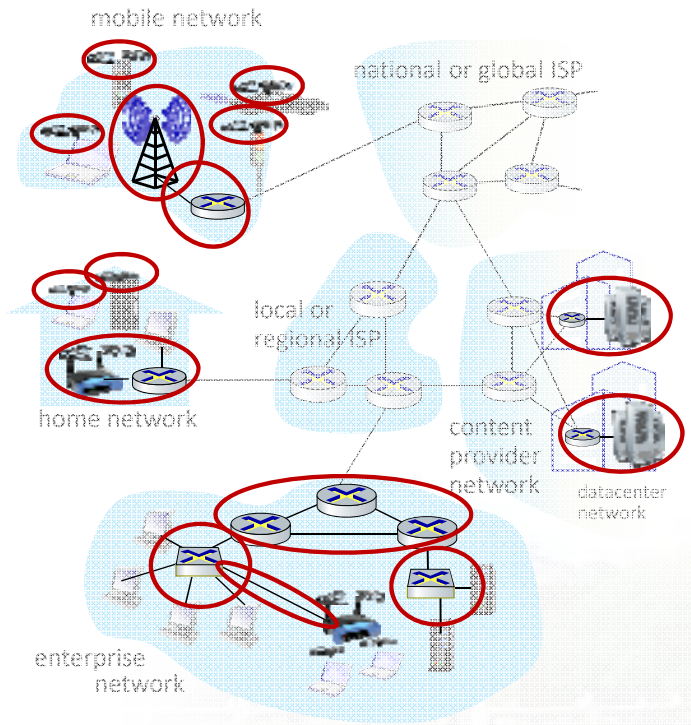
- interconnected routers
- network of networks



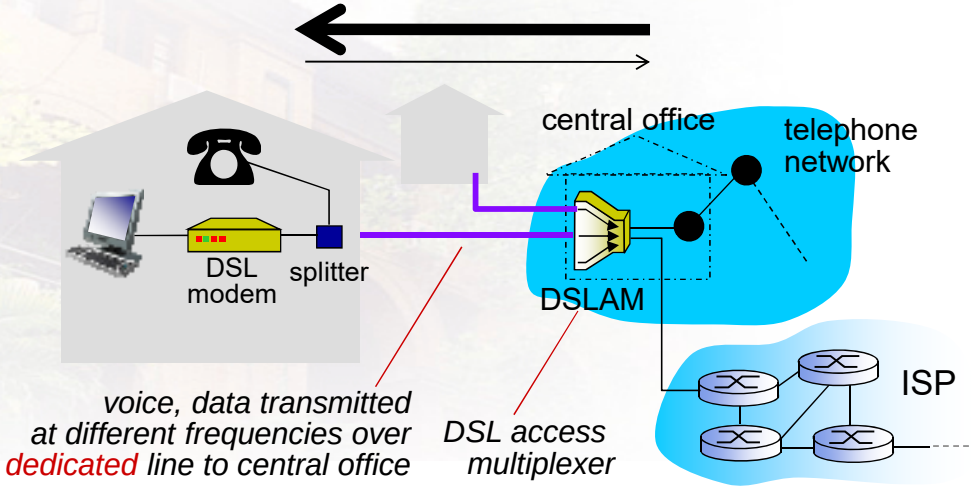
Access networks and physical media

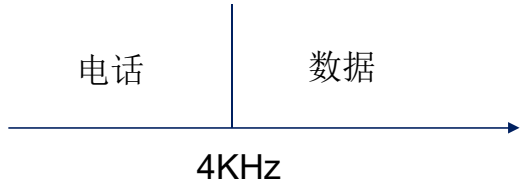
Q: How to connect end systems to edge router?

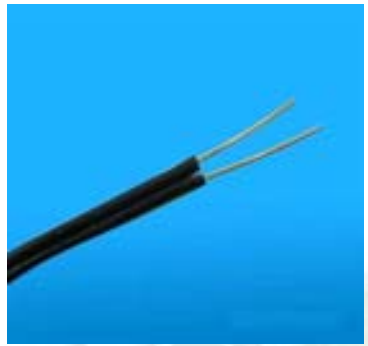
- **residential access nets**
- **institutional access networks (school, company)**
- **mobile access networks (WiFi, 4G/5G)**



Access net: digital subscriber line (DSL)

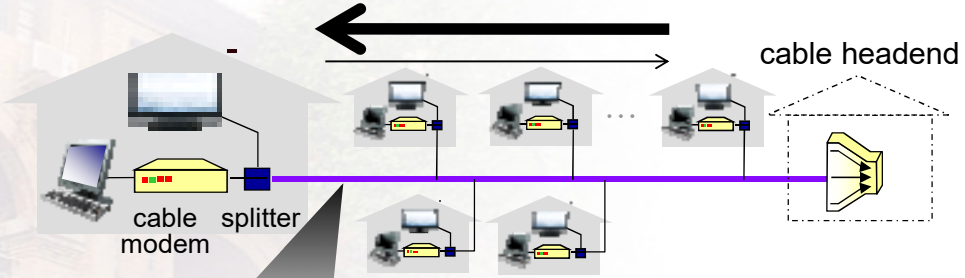


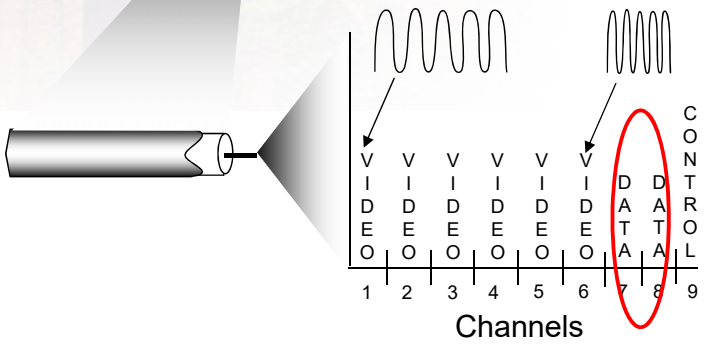




- use *existing* telephone line to central office DSLAM
  - data over DSL phone line goes to Internet
  - voice over DSL phone line goes to telephone net
- < 2.5 Mbps upstream transmission rate (typically < 1 Mbps)
- < 24 Mbps downstream transmission rate (typically < 10 Mbps)

Access net: cable network

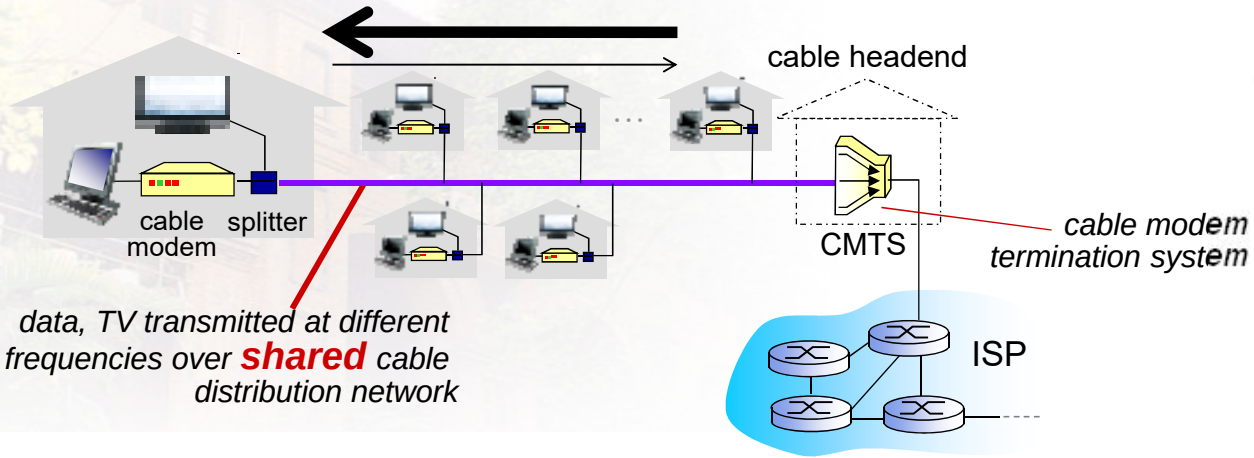






frequency division multiplexing: different channels transmitted in different frequency bands

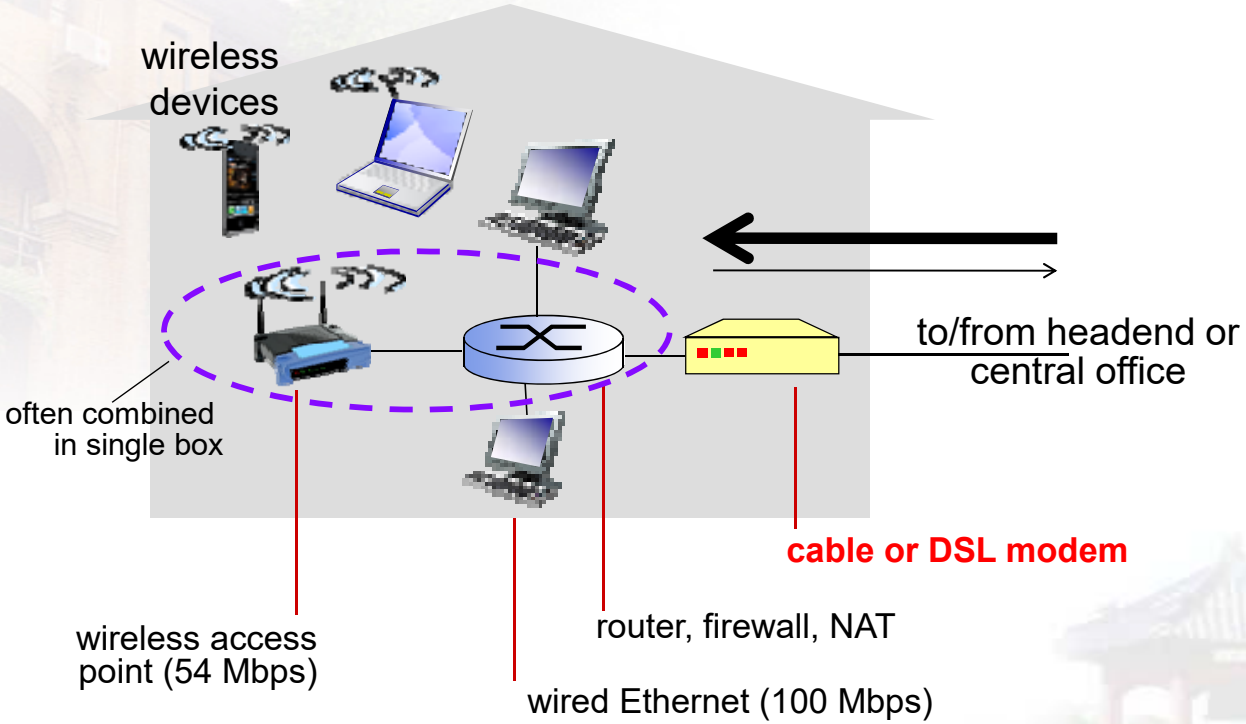
Access net: cable network



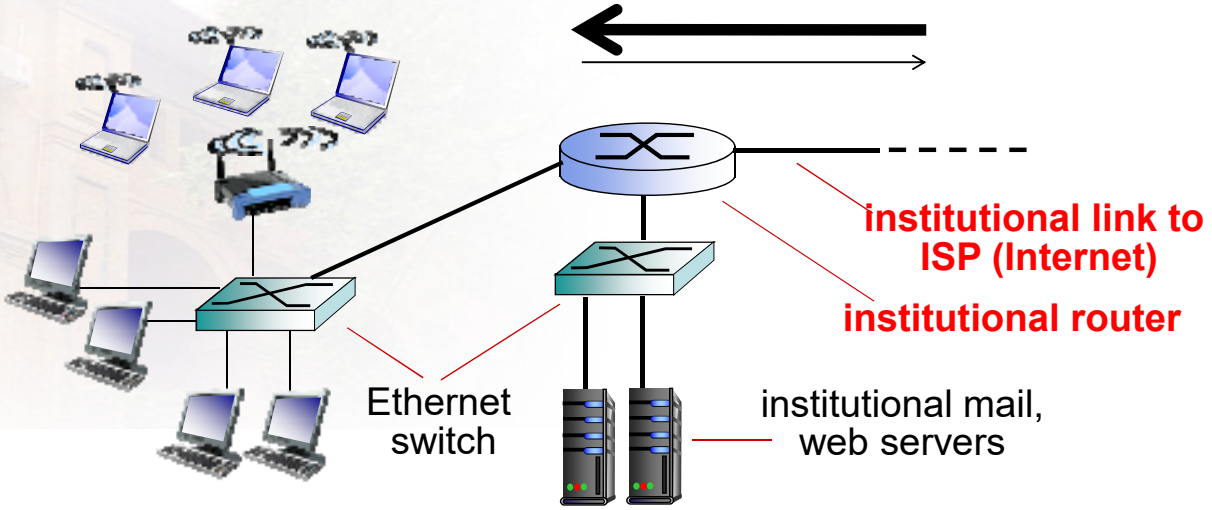


- HFC: hybrid fiber coax
  - asymmetric: up to 30Mbps downstream transmission rate, 2 Mbps upstream transmission rate
- network of cable, fiber attaches homes to ISP router
  - homes ***share access network*** to cable headend
  - unlike DSL, which has dedicated access to central office

Access net: home network



Enterprise access networks (Ethernet)



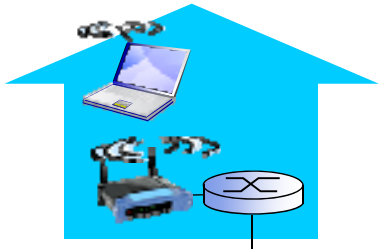
- typically used in companies, universities, etc
- 10 Mbps, 100Mbps, 1Gbps, 10Gbps transmission rates
- today, end systems typically connect into Ethernet switch

Wireless access networks

- **shared *wireless* access network connects end system to router**
  - **via base station aka "access point"**

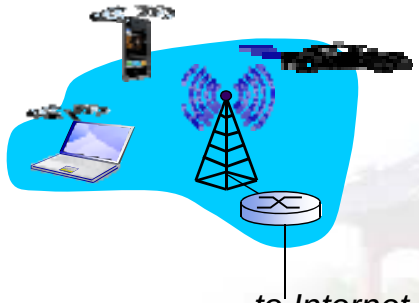
wireless LANs:

- within building (100 ft)
- 802.11b/g (WiFi): 11, 54 Mbps transmission rate



wide-area wireless access

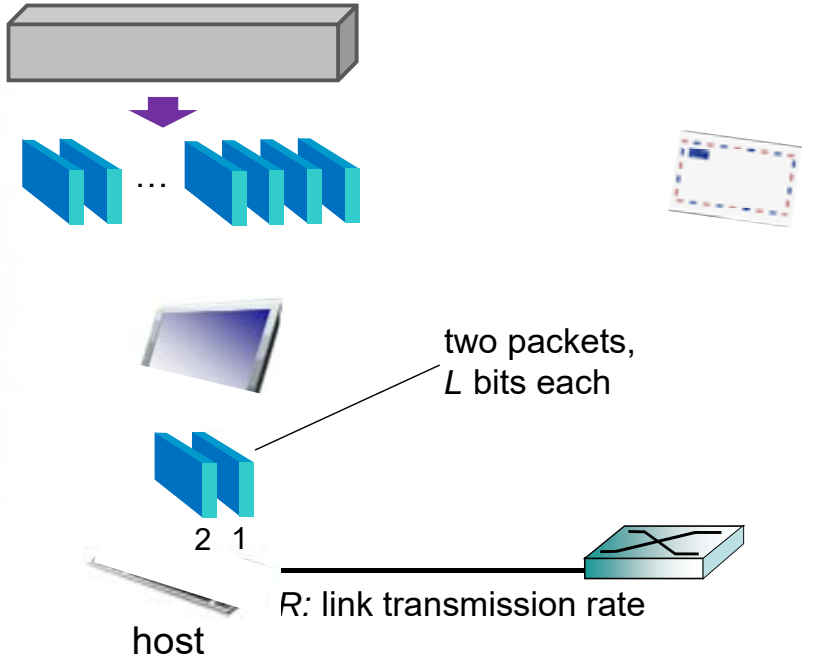
- provided by telco (cellular) operator, 10's km
- between 1 and 10 Mbps
- 3G, 4G, 5G



Host: sends *packets* of data

host sending function:

- **takes application message**
- **breaks into smaller chunks, known as *packets*, of length *L* bits**
- **transmits packet into access network at *transmission rate R***
  - **link transmission rate, aka link *capacity, aka link bandwidth***



$$\text{packet transmission delay} = \text{time needed to transmit } L\text{-bit packet into link} = \frac{L \text{ (bits)}}{R \text{ (bits/sec)}}$$

Physical media

- **bit:** propagates between transmitter/receiver pairs
- **physical link:** what lies between transmitter & receiver
- **guided media:**
  - signals propagate in solid media: copper, fiber, coax
- **unguided media:**
  - signals propagate freely, e.g., radio

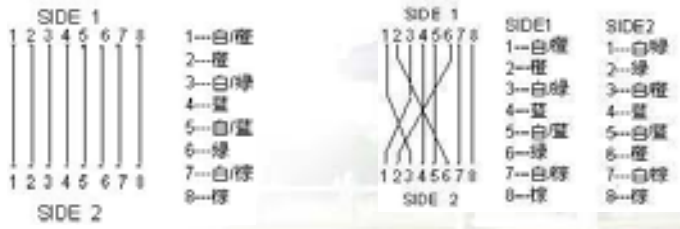
twisted pair (TP)

- two insulated copper wires
  - Category 5: 100 Mbps, 1 Gpbs Ethernet
  - Category 6: 10Gbps









Physical media: coax, fiber

coaxial cable:

- **two concentric copper conductors**
- **bidirectional**
- **broadband:**
  - **multiple channels on cable**
  - **HFC**



fiber optic cable:

- glass fiber carrying light pulses, each pulse a bit
- high-speed operation:
  - high-speed point-to-point transmission (e.g., 10’s-100’s Gpbs transmission rate)
- low error rate:
  - repeaters spaced far apart
  - immune to electromagnetic noise



Links: physical media

Wireless radio

- signal carried in various "bands" in electromagnetic spectrum
- no physical "wire"
- broadcast, "half-duplex" (sender to receiver)
- propagation environment effects:
  - reflection
  - obstruction by objects
  - Interference/noise

Radio link types:

- Wireless LAN (WiFi)
  - 10-100's Mbps; 10's of meters
- wide-area (e.g., 4G cellular)
  - 10's Mbps over ~10 Km
- Bluetooth: cable replacement
  - short distances, limited rates
- terrestrial microwave
  - point-to-point; 45 Mbps channels
- satellite
  - up to 45 Mbps per channel
  - 270 msec end-end delay

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1.2 network edge

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1.3 network core

- packet switching, circuit switching, network structure

1.4 delay, loss, throughput in networks

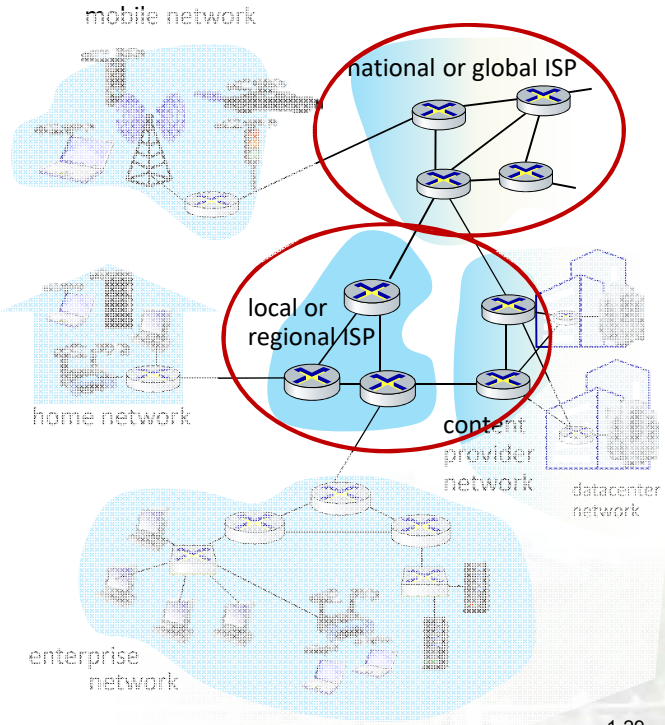
1.5 protocol layers, service models

1.6 networks under attack: security

1.7 history

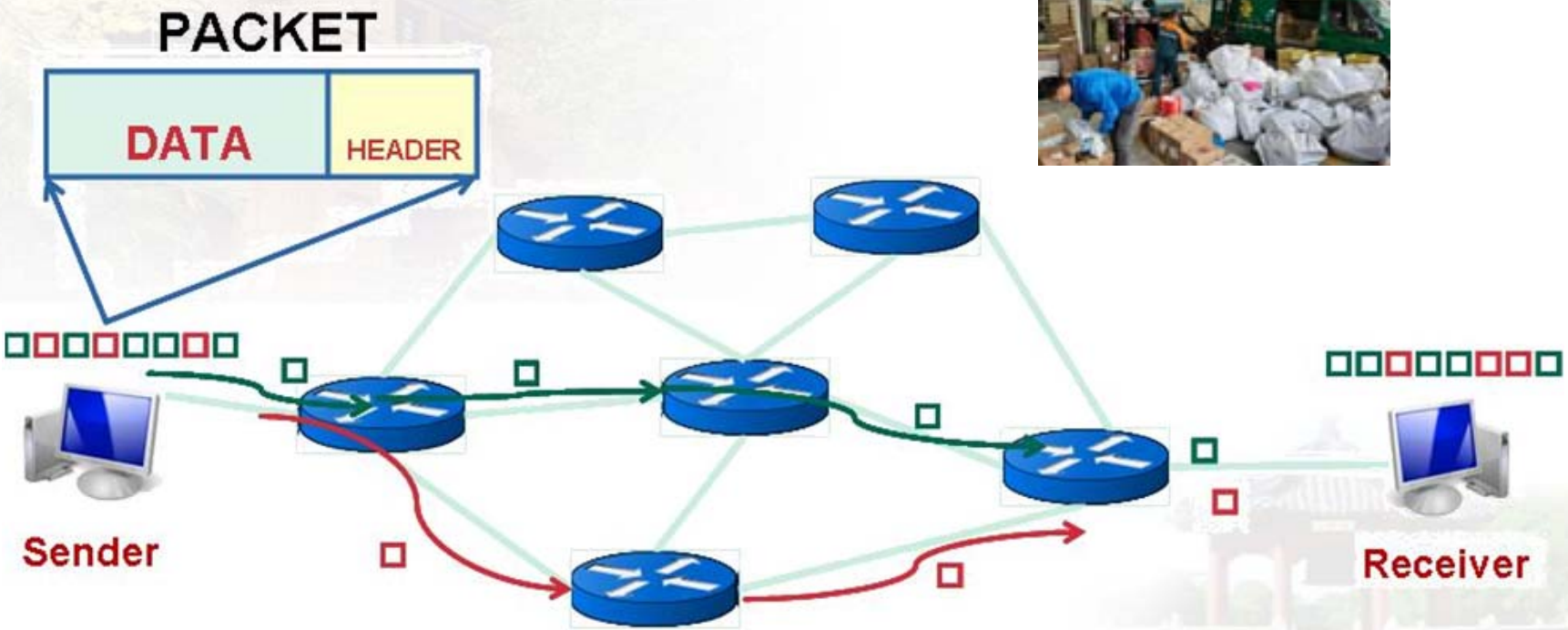
The network core

- mesh of interconnected routers
- packet-switching: hosts break application-layer messages into *packets*
  - forward packets from one router to the next, across links on path from source to destination
  - each packet transmitted at full link capacity

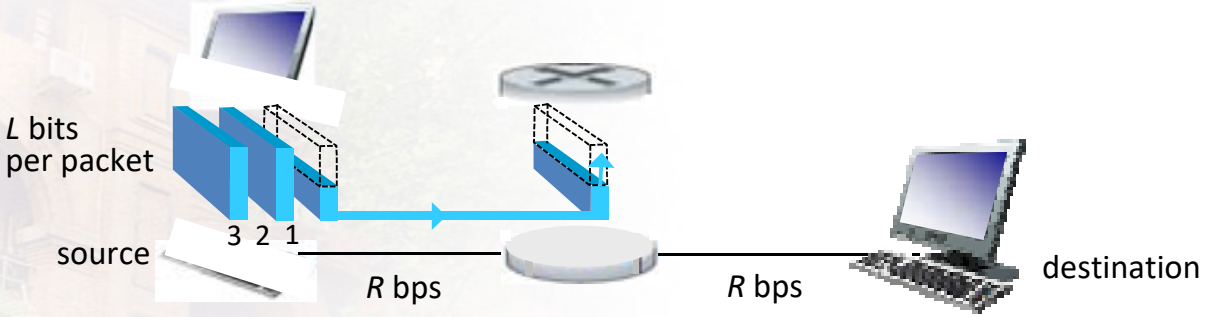


Packet-switching: store-and-forward





Packet-switching: store-and-forward



- takes *L*/*R* seconds to transmit (push out) *L*-bit packet into link at *R* bps
- *store and forward:* entire packet must arrive at router before it can be transmitted on next link

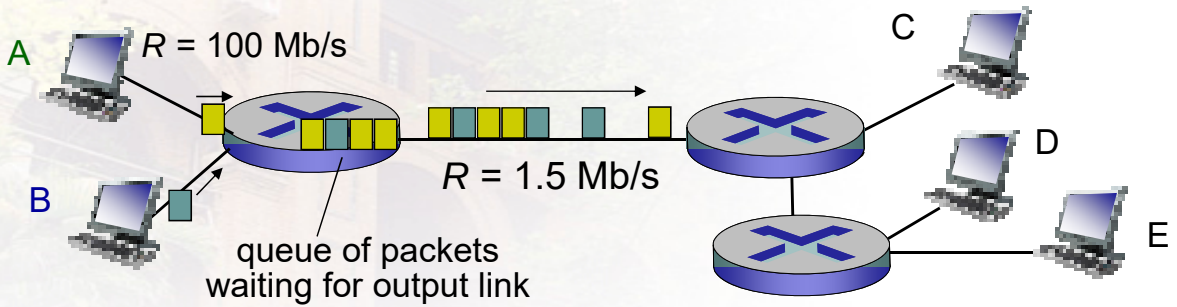
one-hop numerical example:

- *L* = 7.5 Mbits
- *R* = 1.5 Mbps
- one-hop transmission delay = 5 sec

- end-end delay = 2*L*/*R* (assuming zero propagation delay)

more on delay shortly …

Packet Switching: queueing delay, loss





queuing and loss:

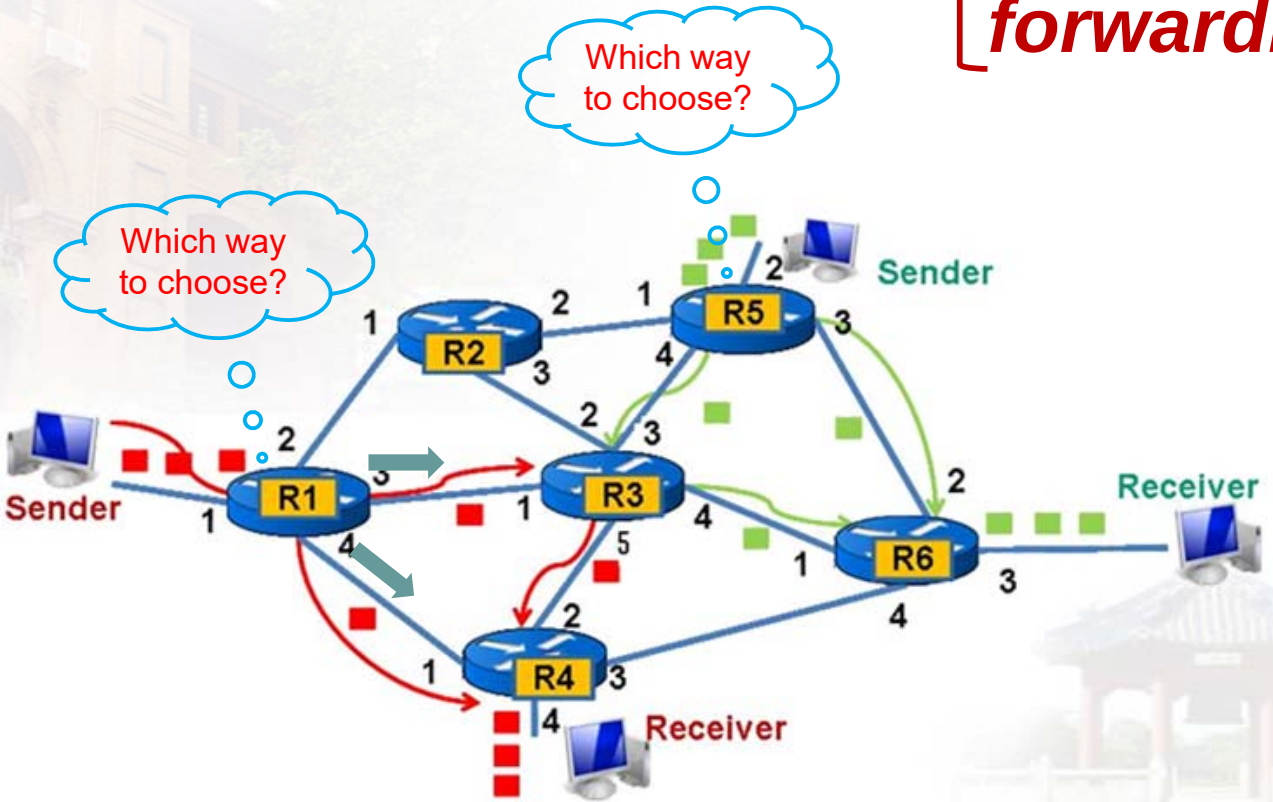
- If arrival rate (in bits) to link exceeds transmission rate of link for a period of time:
  - packets will queue, wait to be transmitted on link
  - packets can be dropped (lost) if memory (buffer) fills up

Queueing occurs when work arrives faster than it can be serviced:

2023.8.30

Two key network-core functions

routing
forwarding



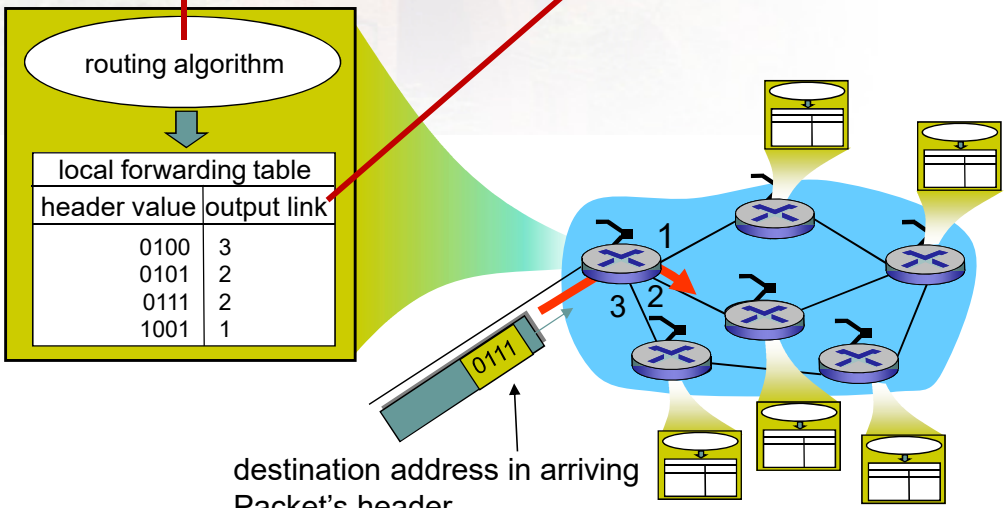
Two key network-core functions

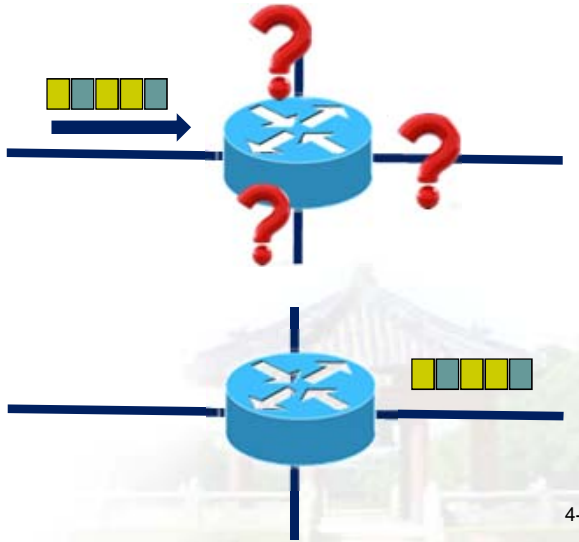
routing: **determines** source-destination **route** taken by packets

- *routing algorithms*

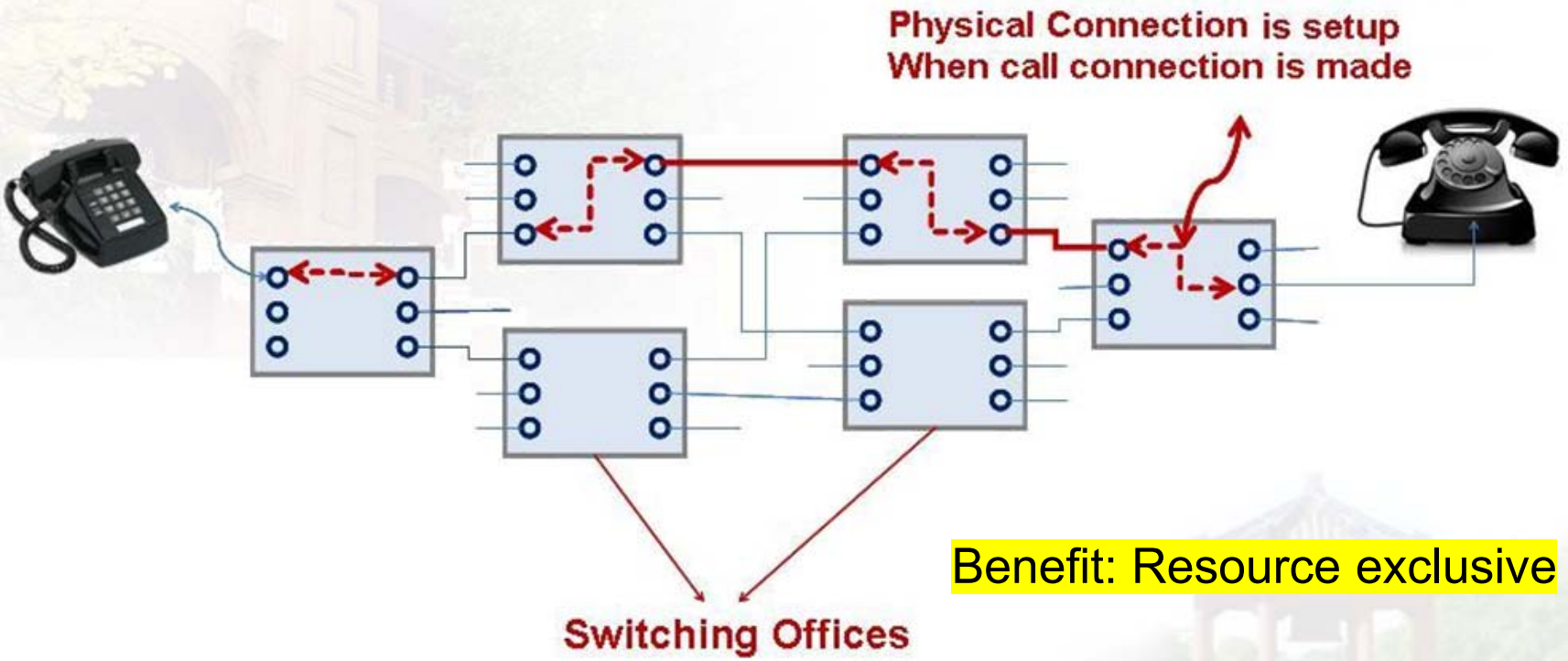
forwarding: **move** **packets from router's input to appropriate router output**

Benefit: Resource sharing





Alternative core: circuit switching

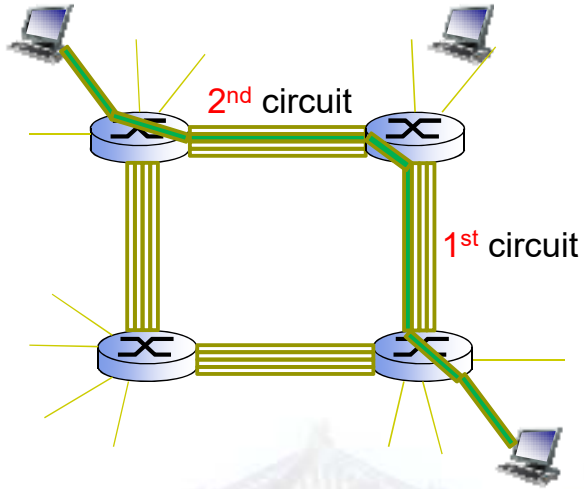


Benefit: Resource exclusive

Alternative core: circuit switching

end-end resources allocated to, reserved for "call" between source & dest:

- In diagram, each link has four circuits.
  - call gets 2nd circuit in top link and 1st circuit in right link.
- dedicated resources: no sharing
  - circuit-like (guaranteed) performance
- circuit segment idle if not used by call *(no sharing)*
- Commonly used in traditional telephone networks



Q: How does circuit switching support multiple users?

Circuit switching: FDM versus TDM

How to allocate channels and circuits

FDM

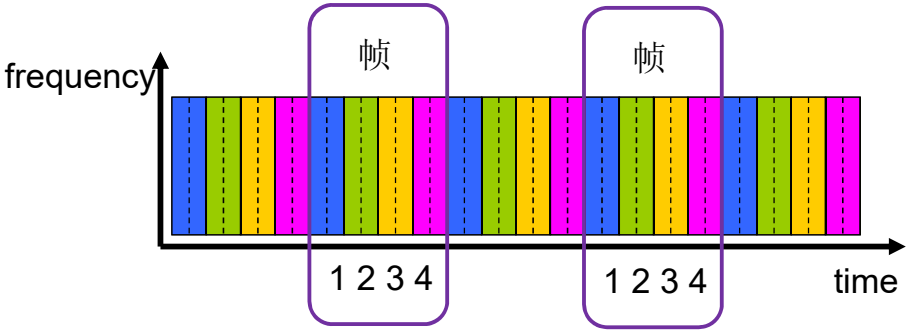


Example:

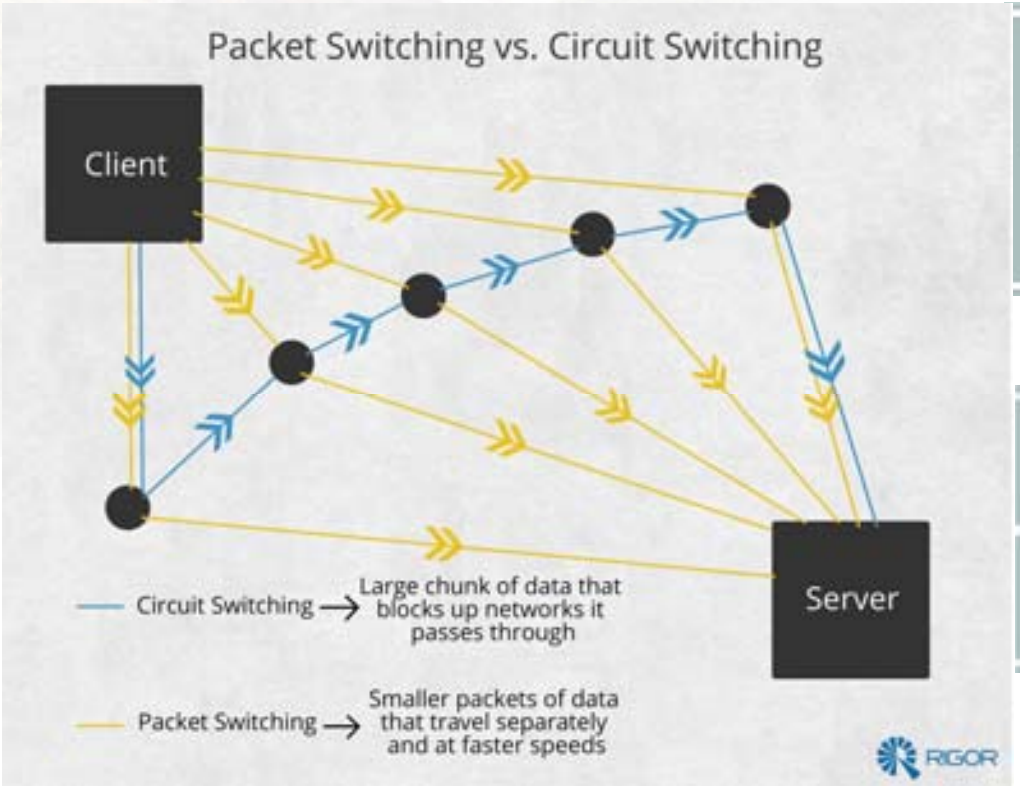
4 users



TDM



Packet switching versus circuit switching





Packet switching



circuit switching

Circuit switching: send data after the path is determined
Packet switching: send data while routing

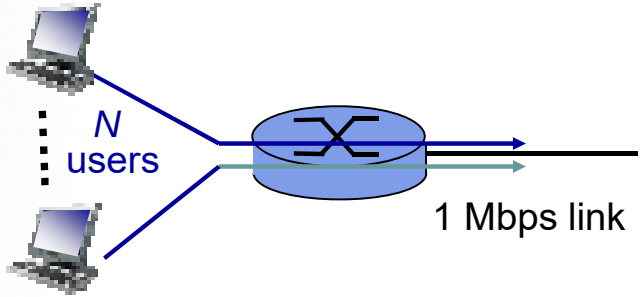
Packet switching versus circuit switching

packet switching allows more users to use network!

example:

- **1 Mb/s link**
- **each user:**
  - **100 kb/s when "active"**
  - **active 10% of time**





- ***circuit-switching:***
  - **10 users 10*100kb/s=1Mb**
- ***packet switching:***
  - **with 35 users, probability > 10 active at same time is less than .0004 ***

Q: how did we get value 0.0004?

(binomial distribution) $P(X=k)=C_n^k p^k (1-p)^{n-k}$

Q: what happens if > 35 users ?

* Check out the online interactive exercises for more examples

Packet switching versus circuit switching

is packet switching a "slam dunk winner?"

- **great for bursty data**
  - **resource sharing**
  - **simpler, no call setup**



- **excessive congestion possible: packet delay and loss**
  - **protocols needed for reliable data transfer, congestion control**
- ***Q:* How to provide circuit-like behavior?**
  - **bandwidth guarantees needed for audio/video apps**
  - **still an unsolved problem**

Q: human analogies of reserved resources (circuit switching) versus on-demand allocation (packet-switching)?

Internet structure: network of networks

- End systems connect to Internet via access ISPs (Internet Service Providers)
  - Residential, company and university ISPs
- Access ISPs in turn must be interconnected.
  - So that any two hosts can send packets to each other
- Resulting network of networks is very complex
  - Evolution was driven by economics and national policies
- Let's take a stepwise approach to describe current Internet structure

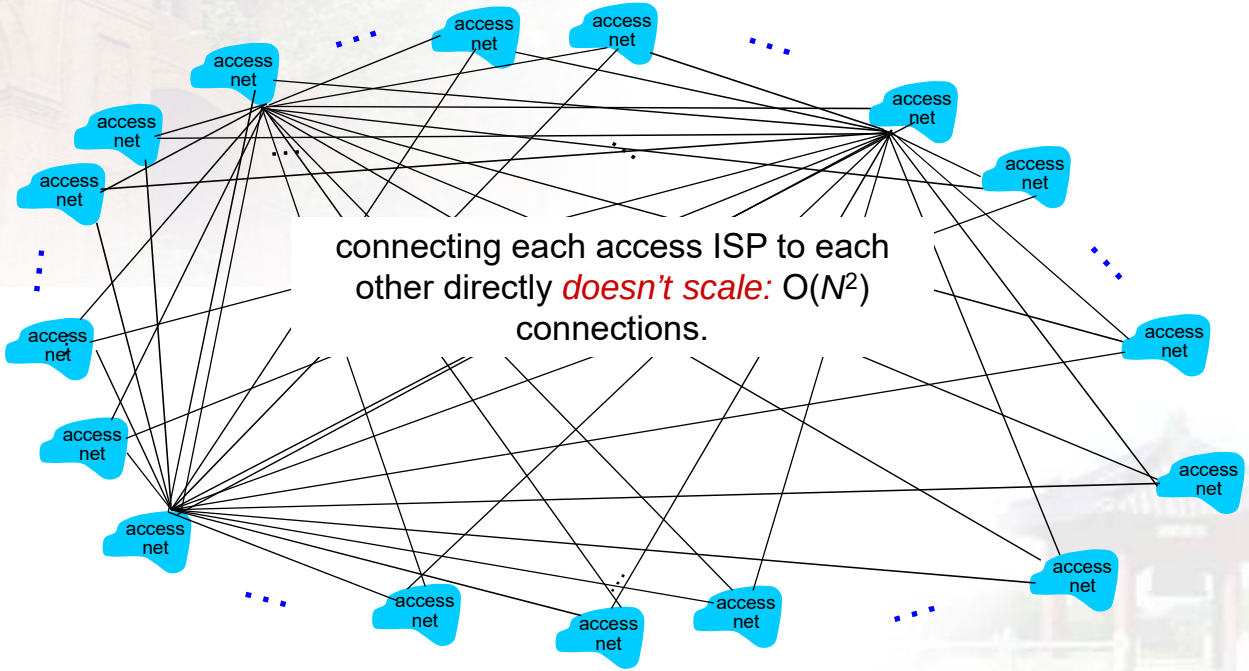
Internet structure: network of networks

Question: **given *millions* of access ISPs, how to connect them together?**



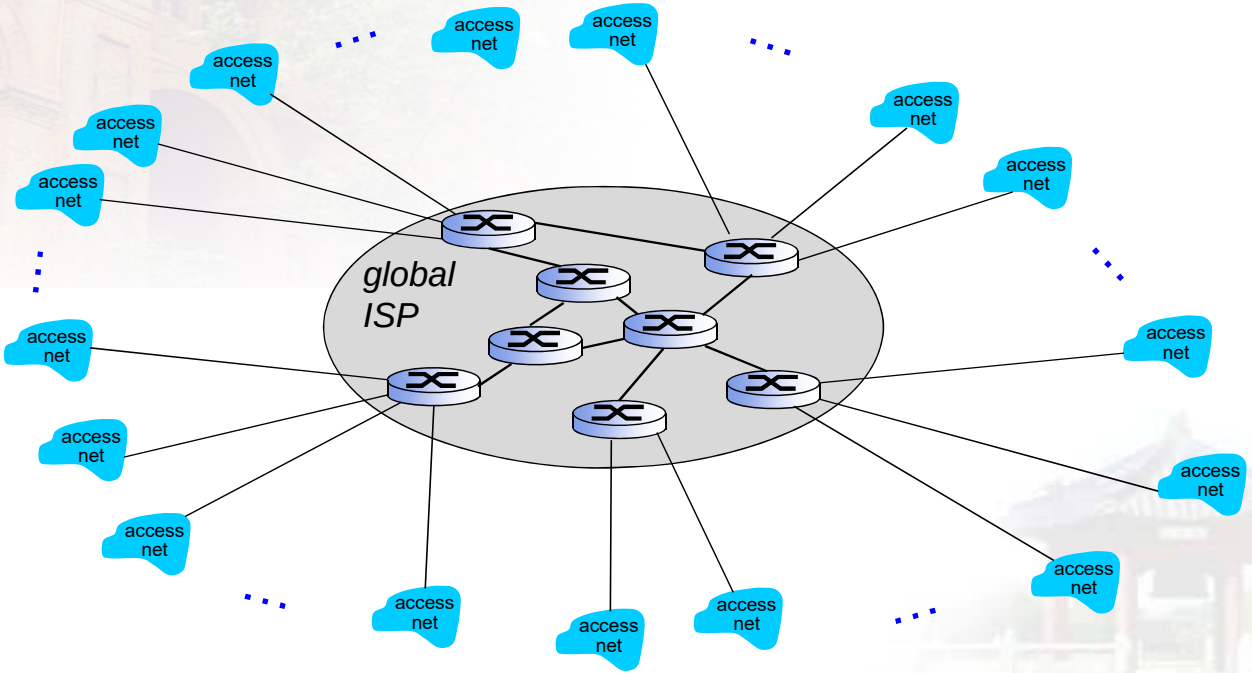
Internet structure: network of networks

***Option:** connect each access ISP to every other access ISP?*



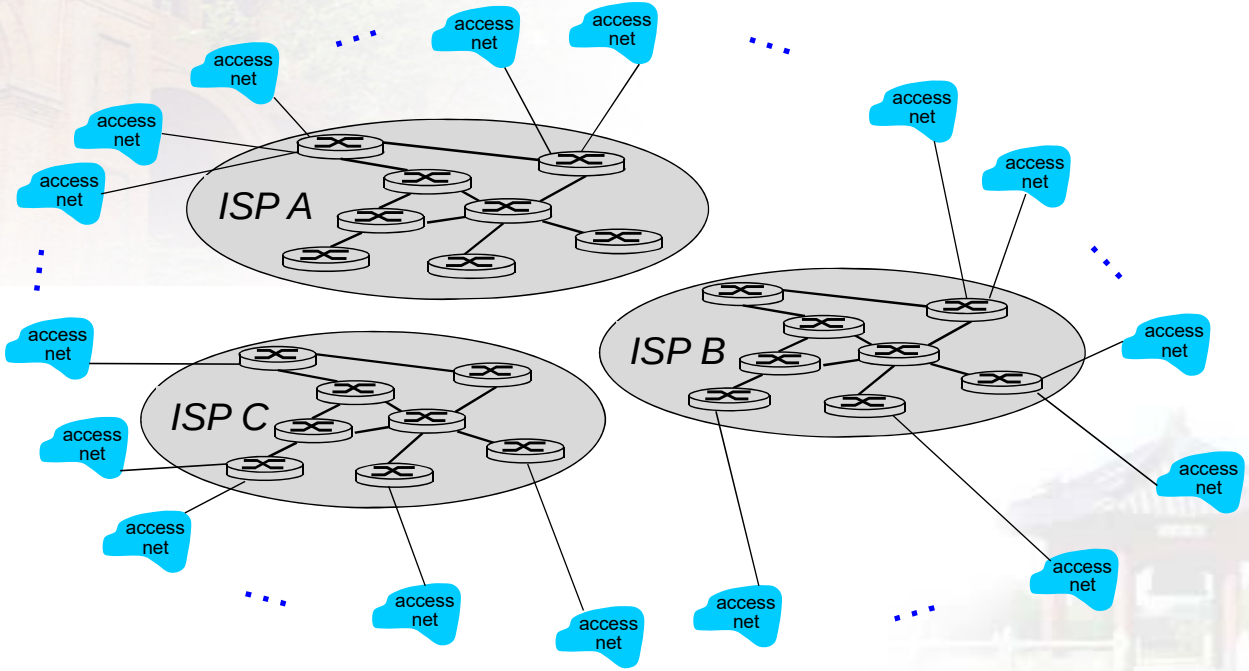
Internet structure: network of networks

Option: connect each access ISP to a global transit ISP? Customer and provider ISPs have economic agreement.



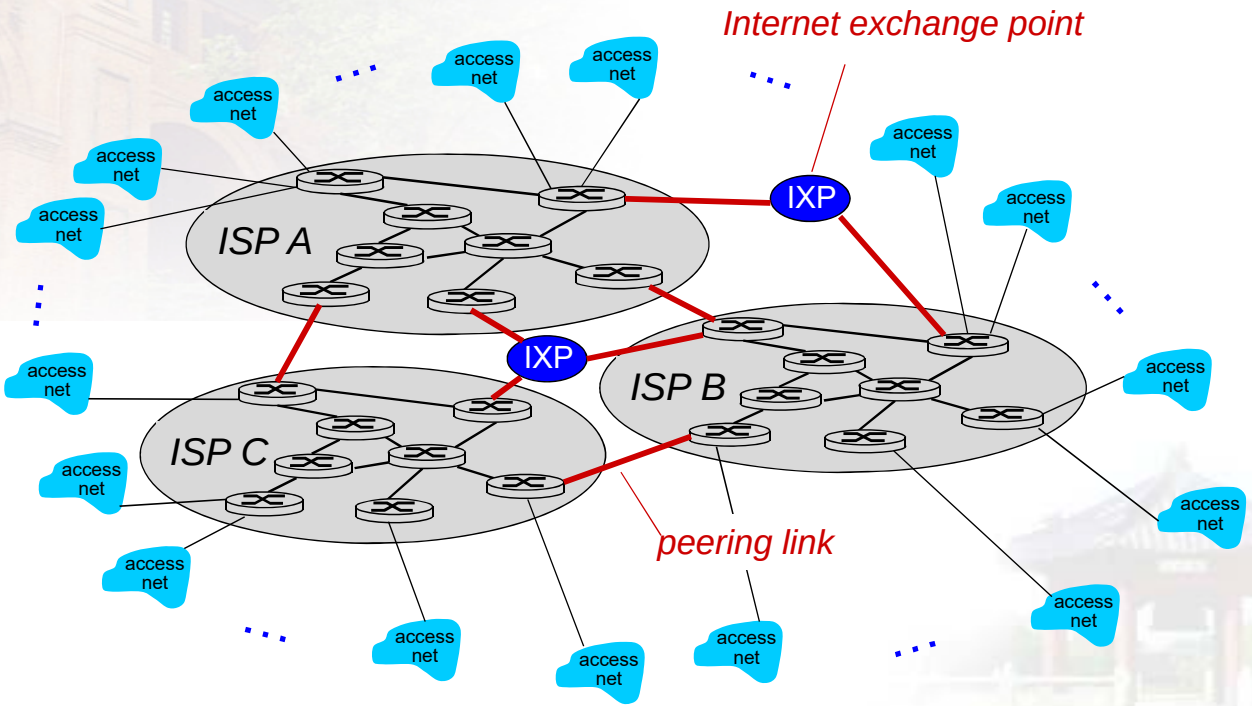
Internet structure: network of networks

But if one global ISP is viable business, there will be competitors ….



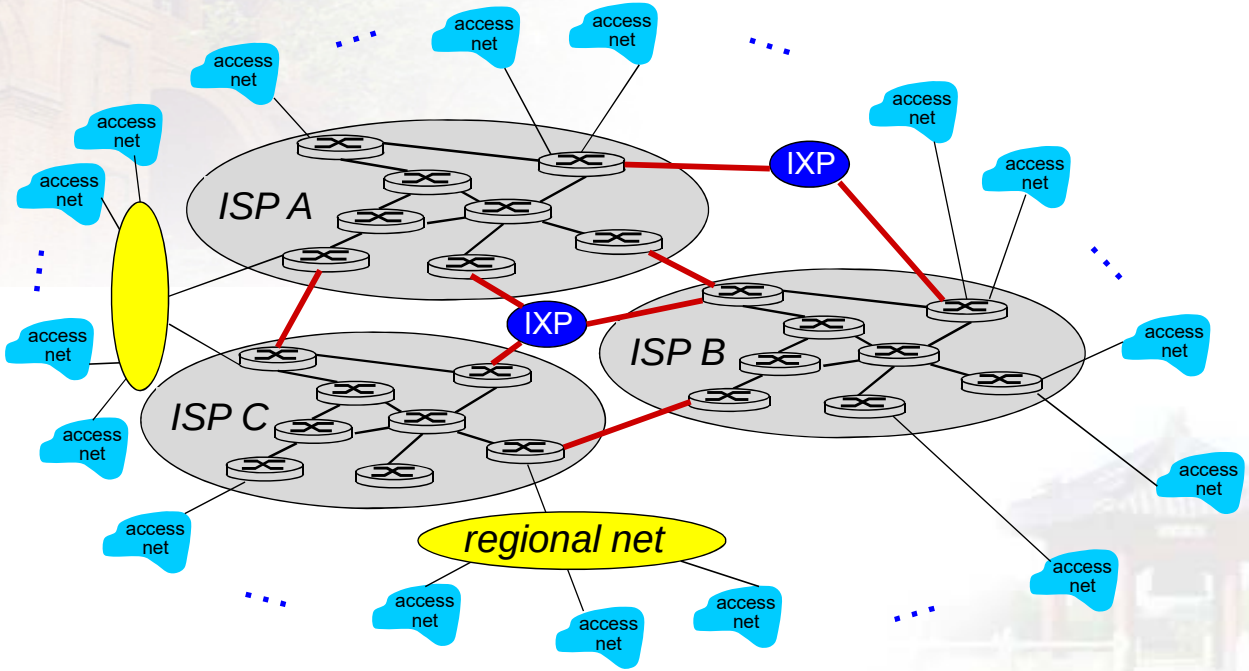
Internet structure: network of networks

But if one global ISP is viable business, there will be competitors …. which must be interconnected



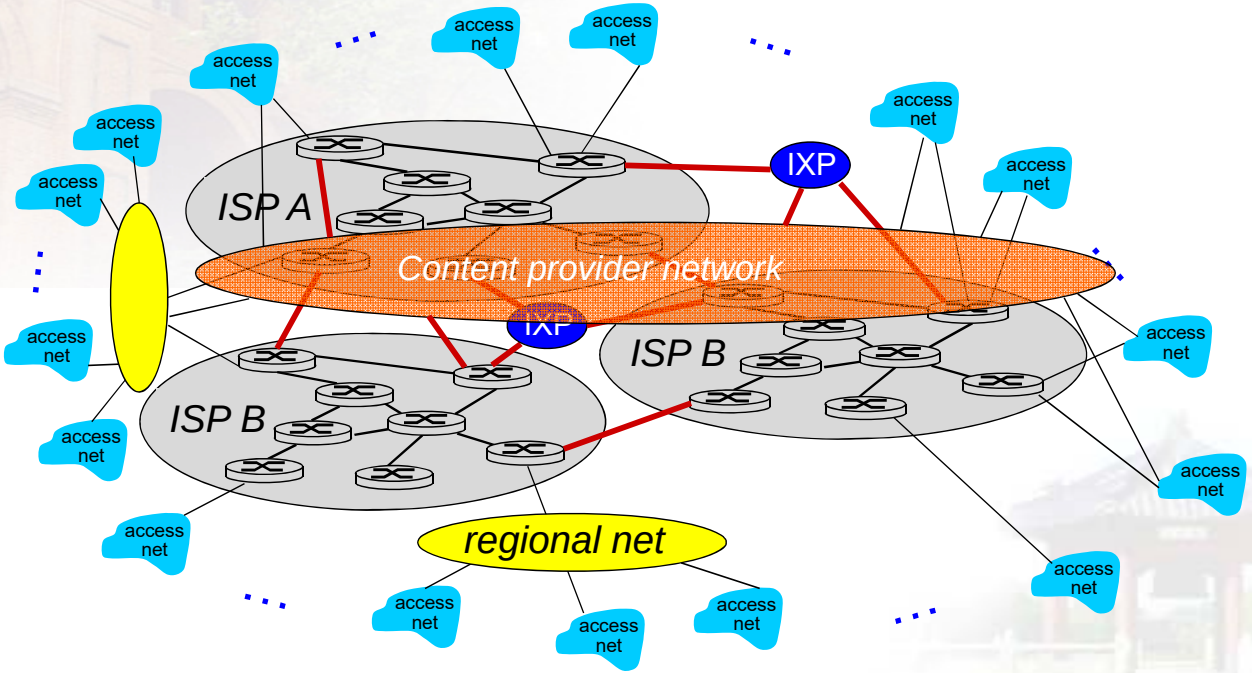
Internet structure: network of networks

… and regional networks may arise to connect access nets to ISPS



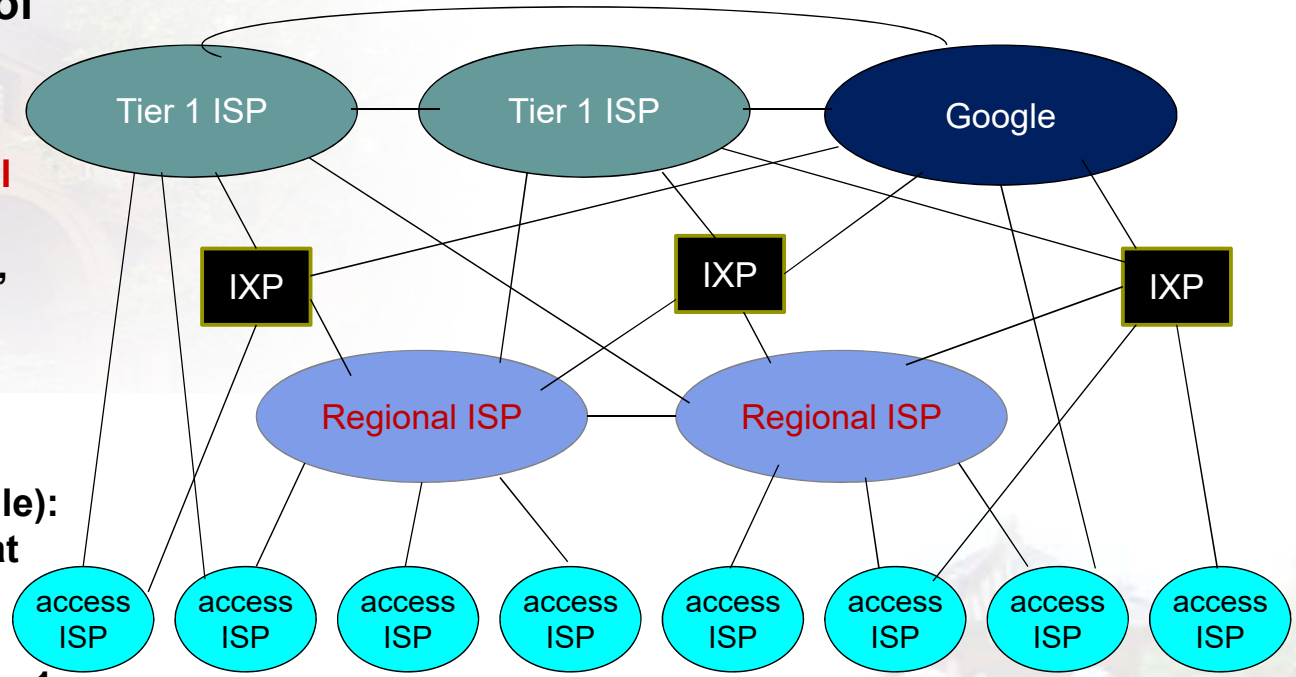
Internet structure: network of networks

… and content provider networks (e.g., Google, Microsoft, Akamai) may run their own network, to bring services, content close to end users



Internet structure: network of networks

- at center: small # of well-connected large networks
  - "tier-1" commercial ISPs (e.g., Level 3, Sprint, AT&T, NTT), national & international coverage
  - content provider network (e.g, Google): private network that connects it data centers to Internet, often bypassing tier-1, regional ISPs



Chapter 1: introduction

1.1 what *is* the Internet?

1.2 network edge

- end systems, access networks, links

1.3 network core

- packet switching, circuit switching, network structure

1.4 delay, loss, throughput in networks

1.5 protocol layers, service models

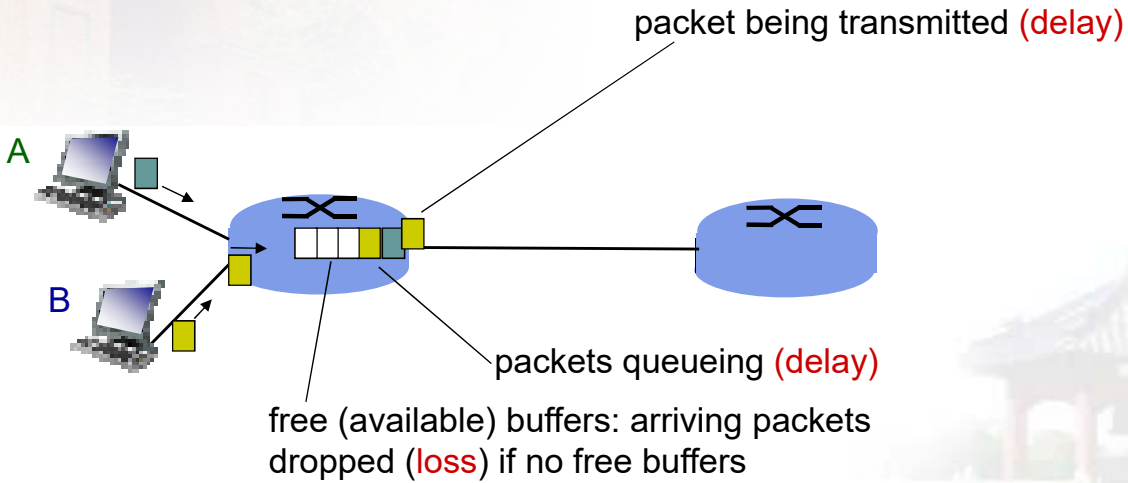
1.6 networks under attack: security

1.7 history

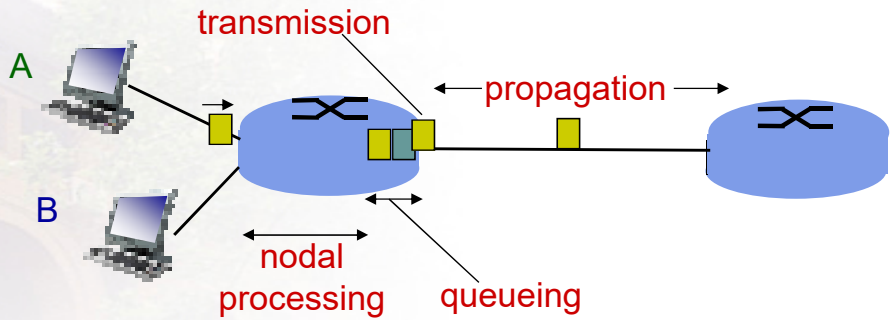
How do loss and delay occur?

packets *queue* in router buffers

- packet arrival rate to link (temporarily) exceeds output link capacity
- packets queue, wait for turn



Four sources of packet delay



$$d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$$

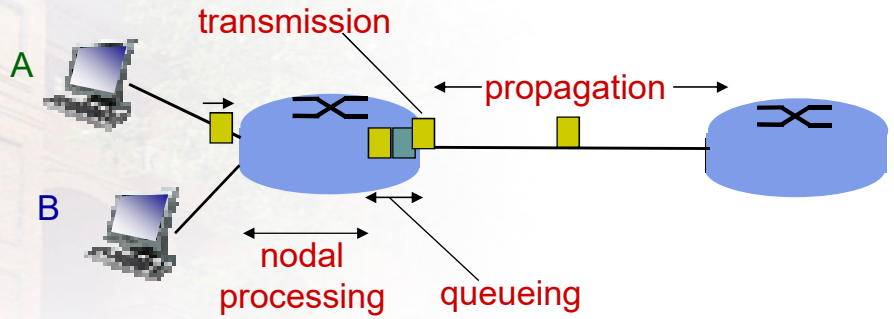
d_{proc}: nodal processing

- **check bit errors**
- **determine output link**
- **typically < msec**

d_{queue}: queueing delay

- time waiting at output link for transmission
- depends on congestion level of router

Four sources of packet delay



$d_{nodal} = d_{proc} + d_{queue} + d_{trans} + d_{prop}$

d_{trans}: transmission delay:

- L: packet length (bits)
- R: link *bandwidth (bps)*
- $d_{trans} = L/R$

d_{prop}: propagation delay:

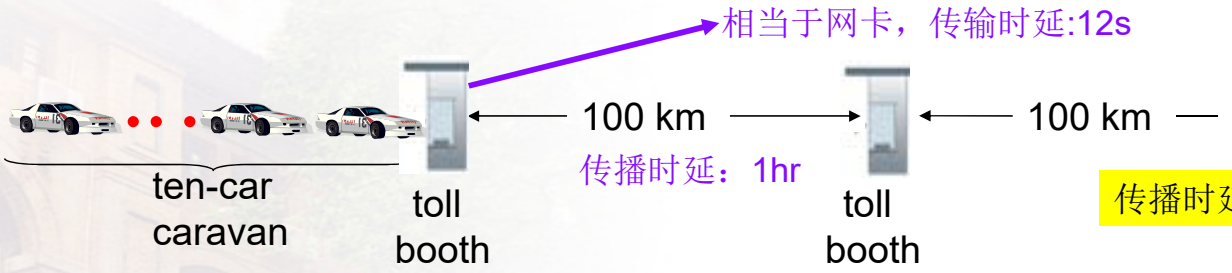
- d: length of physical link
- s: propagation speed in medium (~2×10^8 m/sec)
- $d_{prop} = d/s$

d_{trans} and d_{prop}

Very Very different

* Check out the Java applet for an interactive animation on trans vs. prop delay

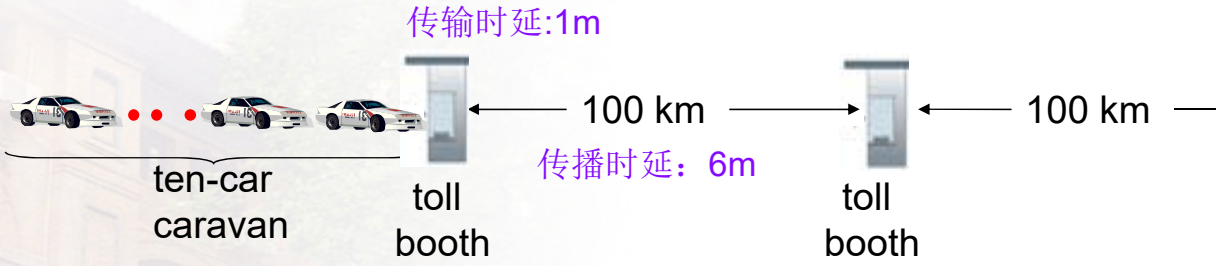
Caravan analogy



- cars "propagate" at 100 km/hr
- toll booth takes 12 sec to service car (bit transmission time)
- car~bit; caravan ~ packet
- *Q:* How long until caravan is lined up before 2nd toll booth?

- time to "push" entire caravan through toll booth onto highway = 12*10 = 120 sec
- time for last car to propagate from 1st to 2nd toll both: 100km/(100km/hr)= 1 hr
- *A:* about 62 minutes

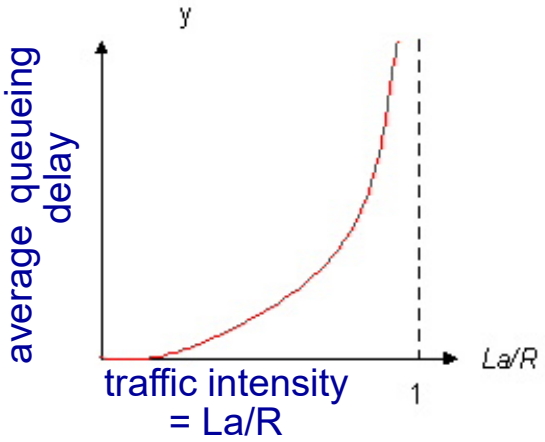
Caravan analogy (more)



- suppose cars now "propagate" at 1000 km/hr
- and suppose toll booth now takes one min to service a car
- *Q:* Will cars arrive to 2nd booth before all cars serviced at first booth?
  - *A: Yes!* after 7 min, 1st car arrives at second booth; three cars still at 1st booth.

Queuing delay (revisited)

- *R:* link bandwidth (bps)
- *L:* packet length (bits)
- a: average packet arrival rate



- *La/R* ~ 0: avg. queuing delay small
- *La/R* → 1: avg. queuing delay large
- *La/R* > 1: more "work" arriving than can be serviced, average delay infinite!



La/R ~ 0

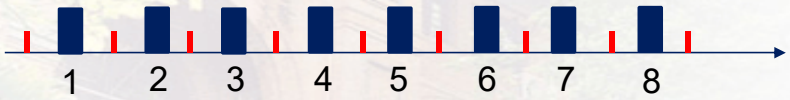


La/R -> 1

* Check out the Java applet for an interactive animation on queuing and loss

- *La/R* → 1: avg. queuing delay large
- *La/R* > 1: more “work” arriving than can be serviced, average delay infinite!

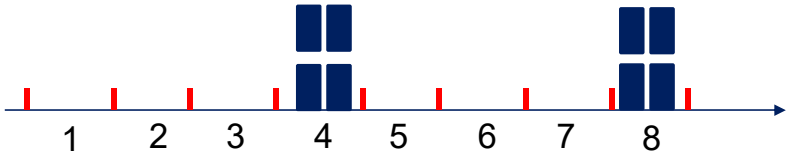
Why?



恒定速率λ到达，LaR=1，满负荷，没有排队



平均速率依然为λ，但是{2，4，6，8}时刻出现排队，各有1次排队

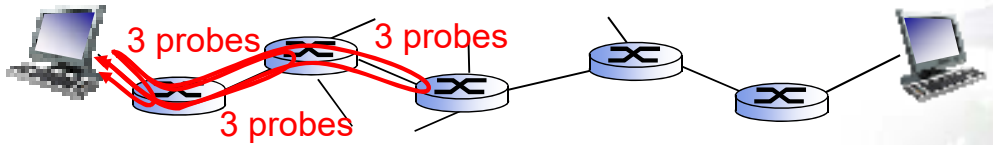


平均速率依然为λ，但是{4，8}时刻出现排队，各有3次排队

原因：分组到达不是恒定速率，具有随机突发性！
理论分析参阅排队论

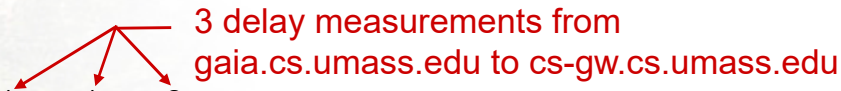
“Real” Internet delays and routes（DIY）

- what do “real” Internet delay & loss look like?
- `Traceroute(win: tracert)` program: provides delay measurement from source to router along end-end Internet path towards destination. For all *i:*
  - sends three UDP packets that will reach router *i* on path towards destination
  - router *i* will return packets to sender
  - sender times interval between transmission and reply.



“Real” Internet delays, routes

traceroute: gaia.cs.umass.edu to www.eurecom.fr




```
1  cs-gw (128.119.240.254)  1 ms  1 ms  2 ms
2  border1-rt-fa5-1-0.gw.umass.edu (128.119.3.145)  1 ms  1 ms  2 ms
3  cht-vbns.gw.umass.edu (128.119.3.130)  6 ms 5 ms 5 ms
4  jn1-at1-0-0-19.wor.vbns.net (204.147.132.129)  16 ms 11 ms 13 ms
5  jn1-so7-0-0-0.wae.vbns.net (204.147.136.136)  21 ms 18 ms 18 ms
6  abilene-vbns.abilene.ucaid.edu (198.32.11.9)  22 ms  18 ms  22 ms
7  nycm-wash.abilene.ucaid.edu (198.32.8.46)  22 ms  22 ms  22 ms
8  62.40.103.253 (62.40.103.253)  104 ms 109 ms 106 ms
9  de2-1.de1.de.geant.net (62.40.96.129)  109 ms 102 ms 104 ms
10  de.fr1.fr.geant.net (62.40.96.50)  113 ms 121 ms 114 ms
11  renater-gw.fr1.fr.geant.net (62.40.103.54)  112 ms  114 ms  112 ms
12  nio-n2.cssi.renater.fr (193.51.206.13)  111 ms  114 ms  116 ms
13  nice.cssi.renater.fr (195.220.98.102)  123 ms  125 ms  124 ms
14  r3t2-nice.cssi.renater.fr (195.220.98.110)  126 ms  126 ms  124 ms
15  eurecom-valbonne.r3t2.ft.net (193.48.50.54)  135 ms  128 ms  133 ms
16  194.214.211.25 (194.214.211.25)  126 ms  128 ms  126 ms
17  * * *
18  * * *
19  fantasia.eurecom.fr (193.55.113.142)  132 ms  128 ms  136 ms
```

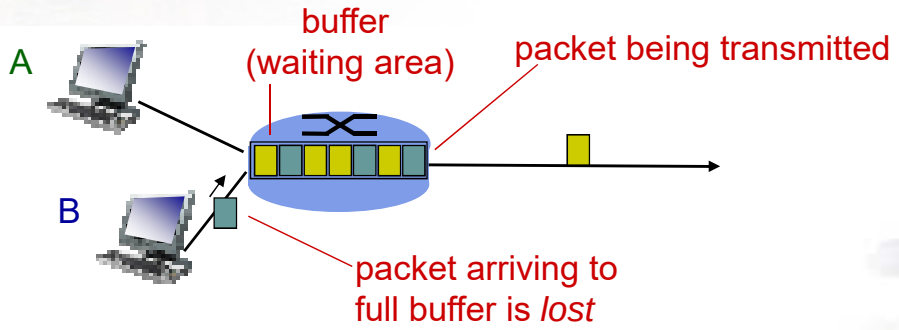
trans-oceanic link

* means no response (probe lost, router not replying)

* Do some traceroutes from exotic countries at www.traceroute.org

Packet loss

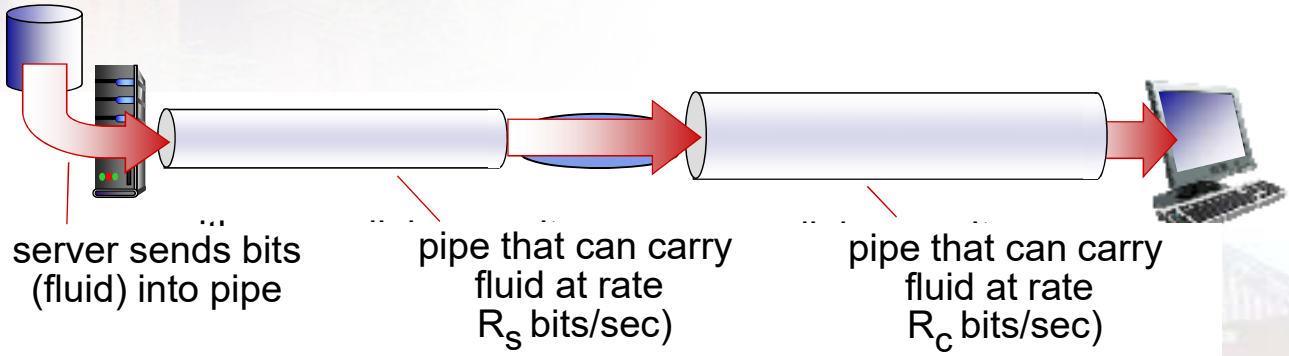
- queue (aka buffer) preceding link in buffer has finite capacity
- packet arriving to full queue dropped (aka lost)
- lost packet **may be retransmitted** by previous node, by source end system, **or not at all**



* Check out the Java applet for an interactive animation on queuing and loss

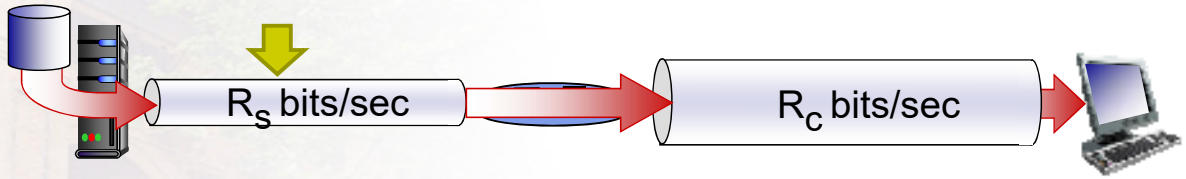
Throughput

- ***throughput:*** **rate (bits/time unit) at which bits transferred between sender/receiver**
  - ***instantaneous:*** **rate at given point in time**
  - ***average:*** **rate over longer period of time**

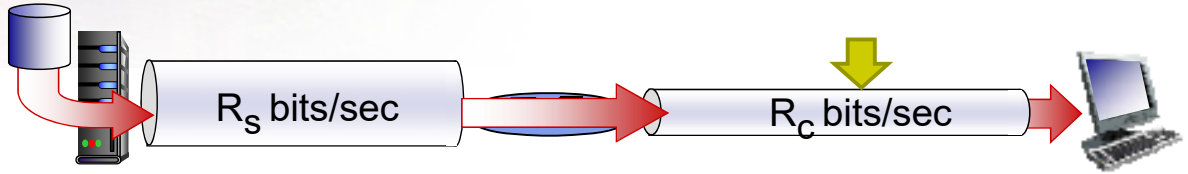


Throughput (more)

- $R_s < R_c$ **What is average end-end throughput?**



  - $R_s > R_c$ What is average end-end throughput?

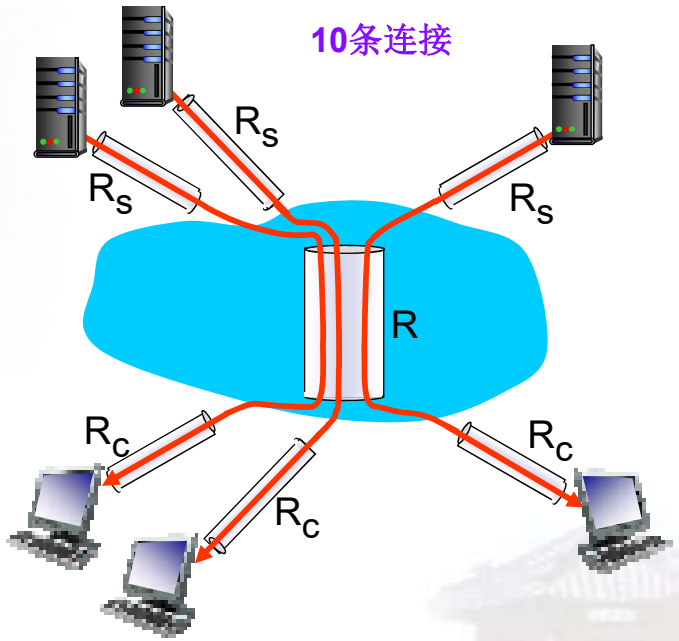


bottleneck link

link on end-end path that constrains end-end throughput

Throughput: Internet scenario

- per-connection end-end throughput: $\min(R_c, R_s, R/10)$
- in practice: R_c or R_s is often bottleneck



10 connections (fairly) share backbone bottleneck link R bits/sec

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1.5 protocol layers, service models

1.6 networks under attack: security

1.7 history

Protocol "layers"

Networks are complex, with many "pieces":

- hosts
- routers
- links of various media
- applications
- protocols
- hardware, software

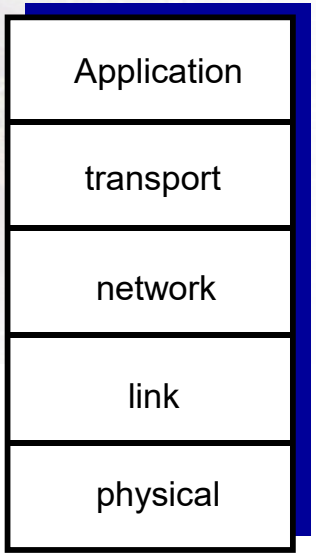


Question:

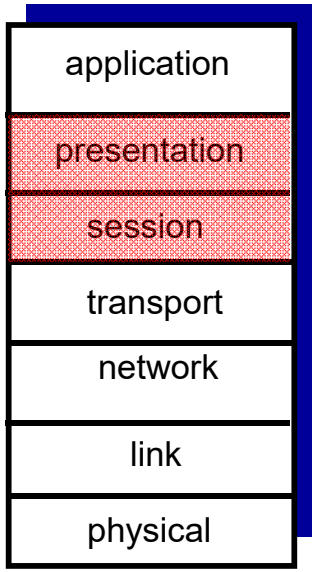
is there any hope of *organizing* structure of network?

…. or at least our discussion of networks?

Protocol “layers”

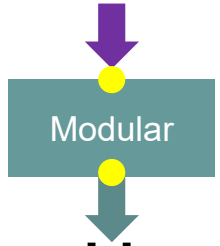


Internet protocol stack



ISO/OSI reference model

Why layering?



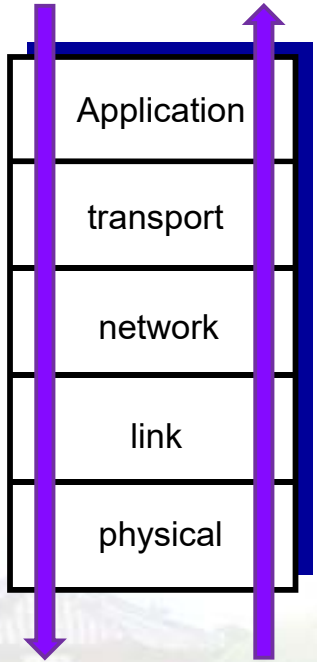
dealing with complex systems:

- **explicit structure allows identification, relationship of complex system's pieces**
  - **layered *reference model* for discussion**
- **modularization eases maintenance, updating of system**
  - **change of implementation of layer' s service transparent to rest of system**
  - **e.g., change in gate procedure doesn't affect rest of system**
- **layering considered harmful?**



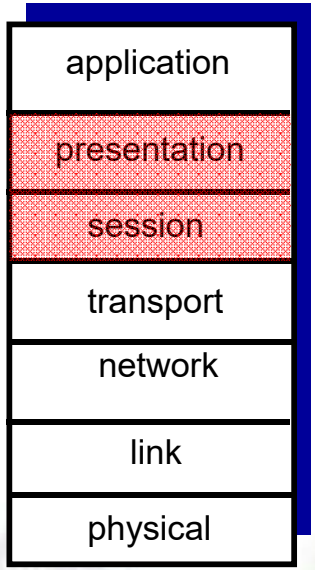
Internet protocol stack

- *application:* supporting network applications
  - FTP, SMTP, HTTP
- *transport:* process-process data transfer
  - TCP, UDP
- *network:* routing of datagrams from source to destination
  - IP, routing protocols
- *link:* data transfer between neighboring network elements
  - Ethernet, 802.111 (WiFi), PPP
- *physical:* bits "on the wire"

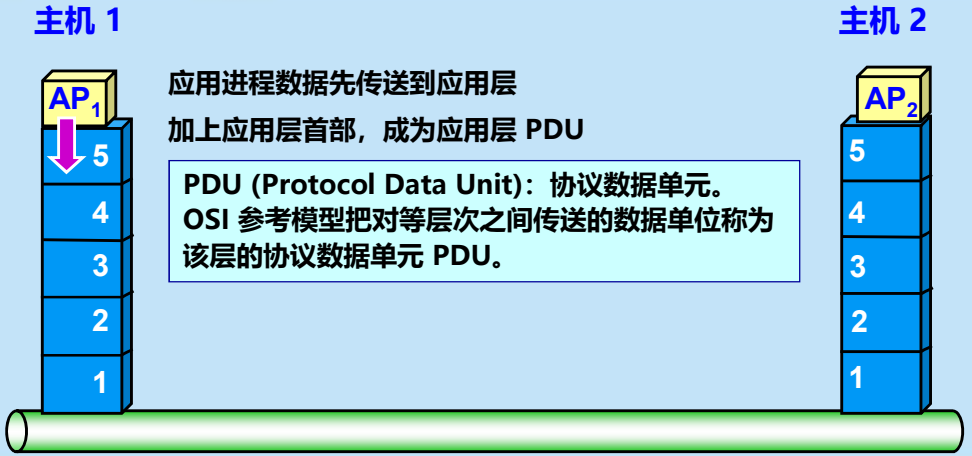


ISO/OSI reference model

- *presentation:* allow applications to interpret meaning of data, e.g., encryption, compression, machine-specific conventions
- *session:* synchronization, check pointing, recovery of data exchange
- Internet stack "missing" these layers!
  - these services, *if needed,* must be implemented in application
  - needed?



主机 1 向主机 2 发送数据

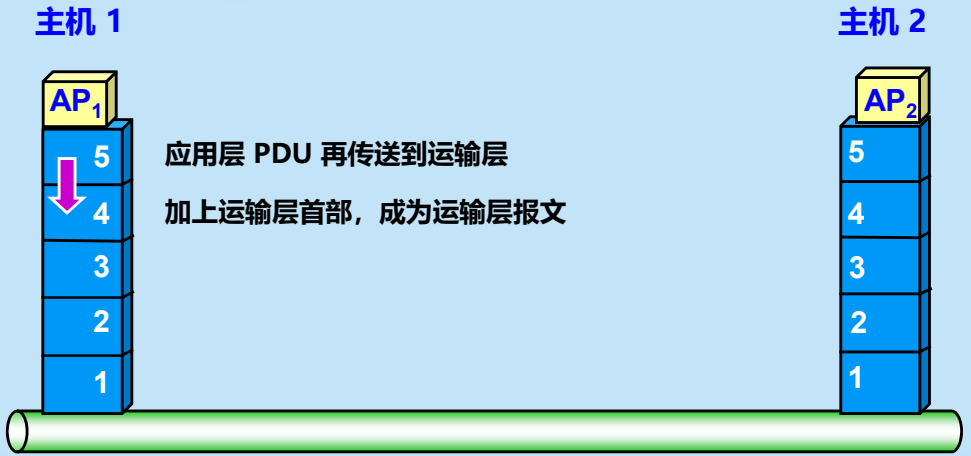


应用进程数据先传送到应用层

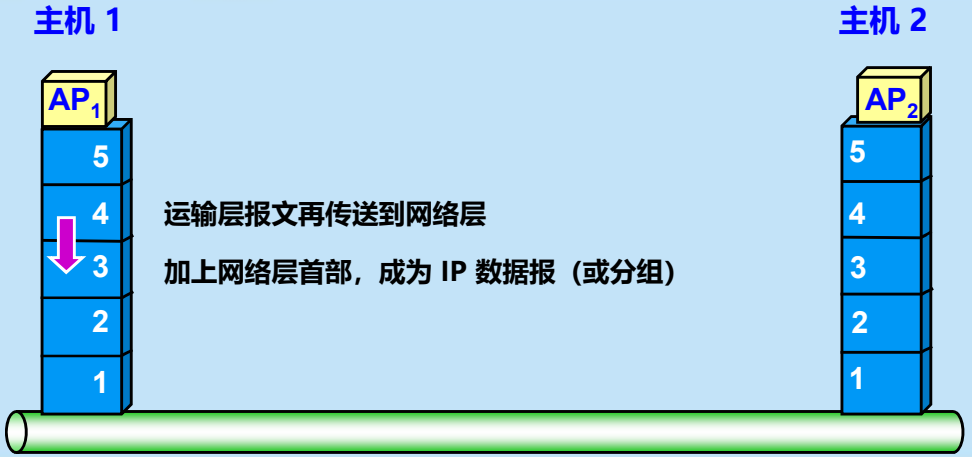
加上应用层首部，成为应用层 PDU

PDU (Protocol Data Unit)：协议数据单元。
OSI 参考模型把对等层次之间传送的数据单位称为该层的协议数据单元 PDU。

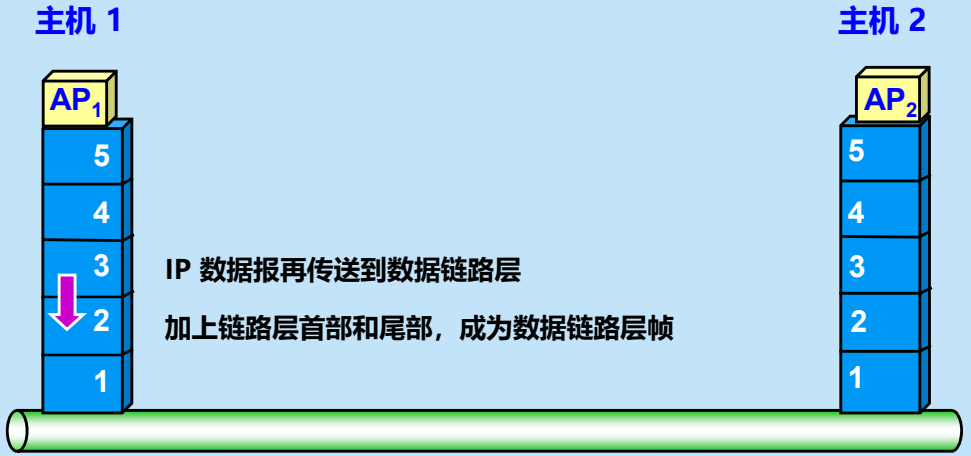
主机 1 向主机 2 发送数据



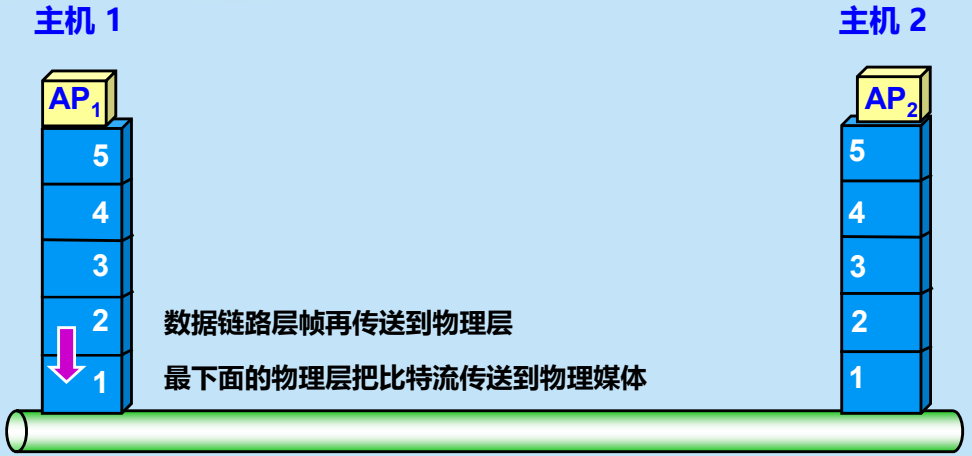
主机 1 向主机 2 发送数据



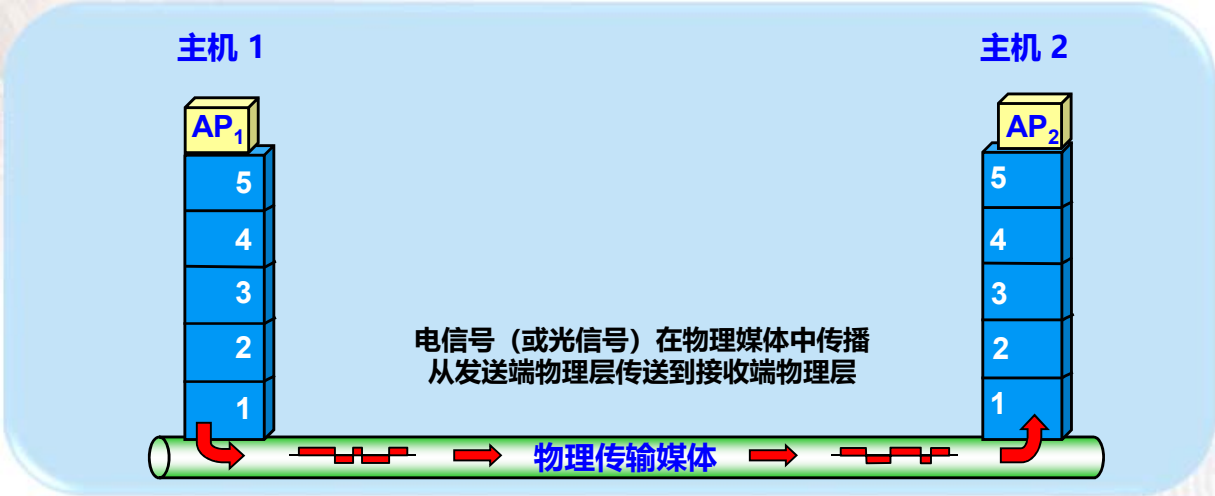
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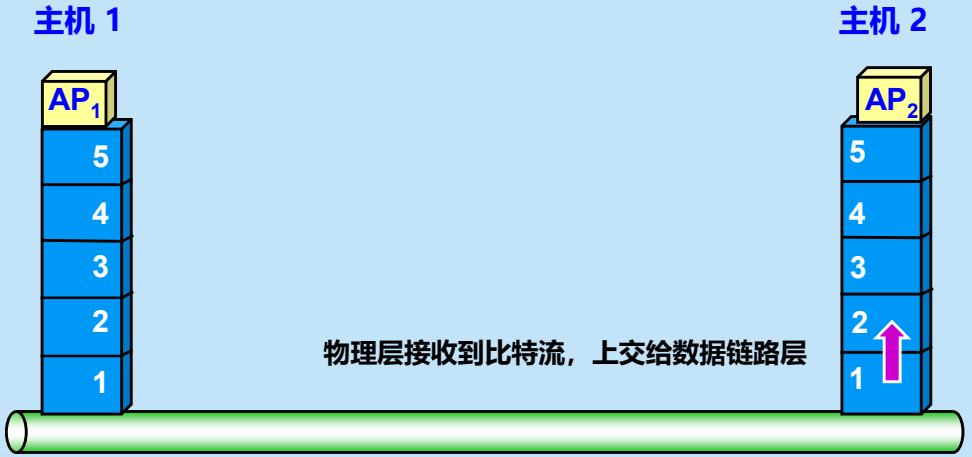
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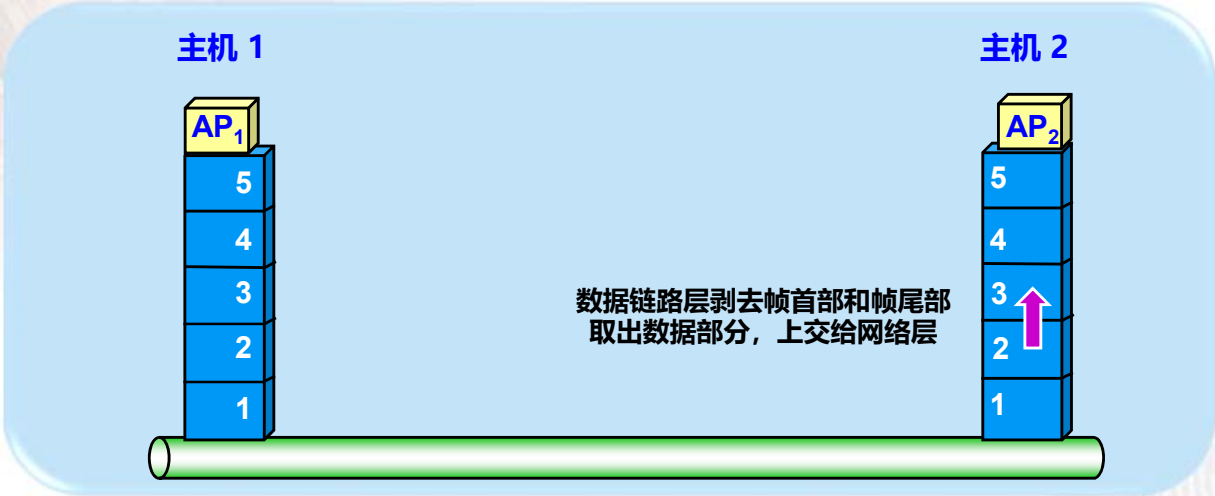
主机 1 向主机 2 发送数据



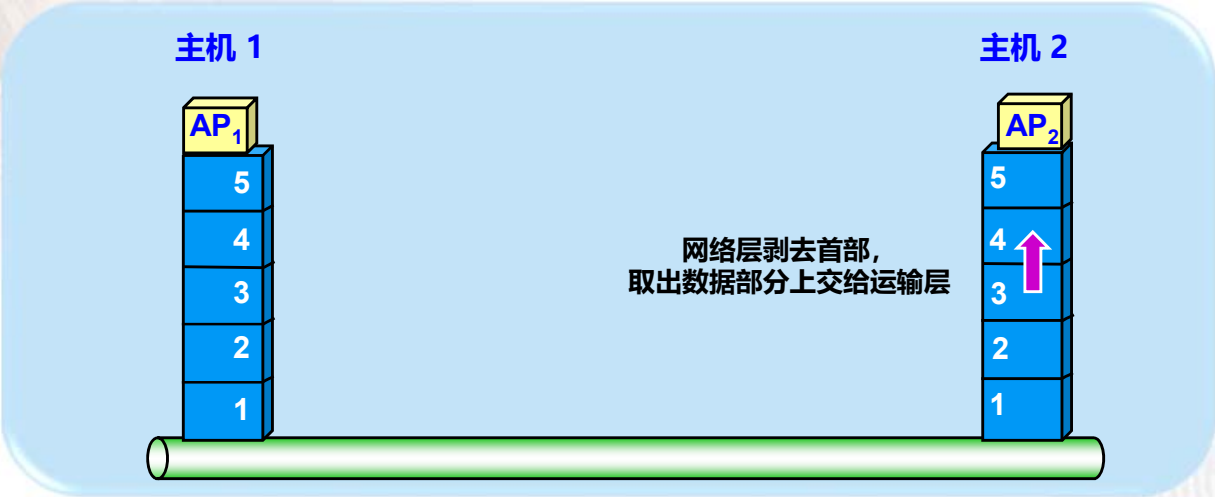
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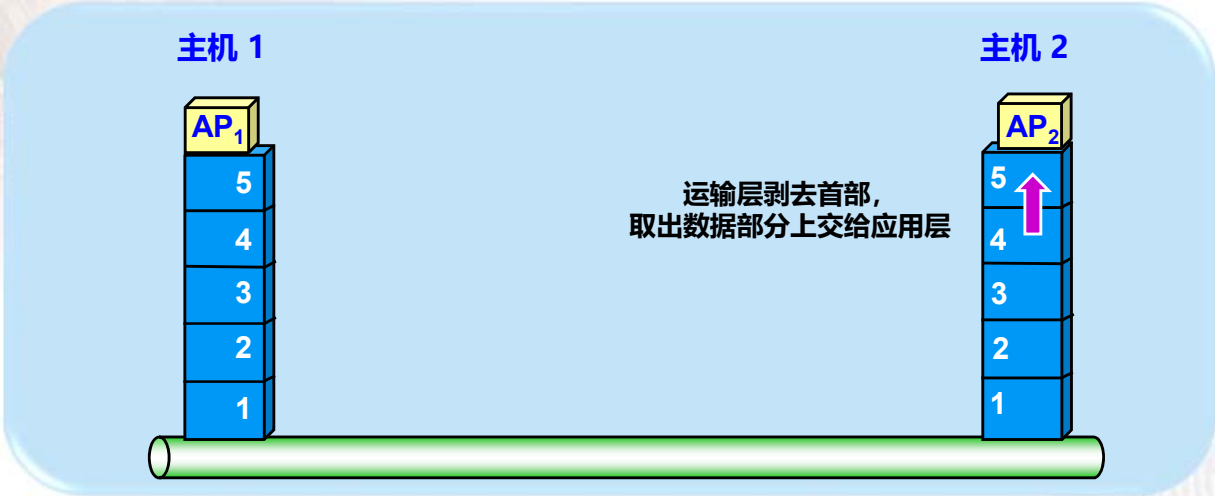
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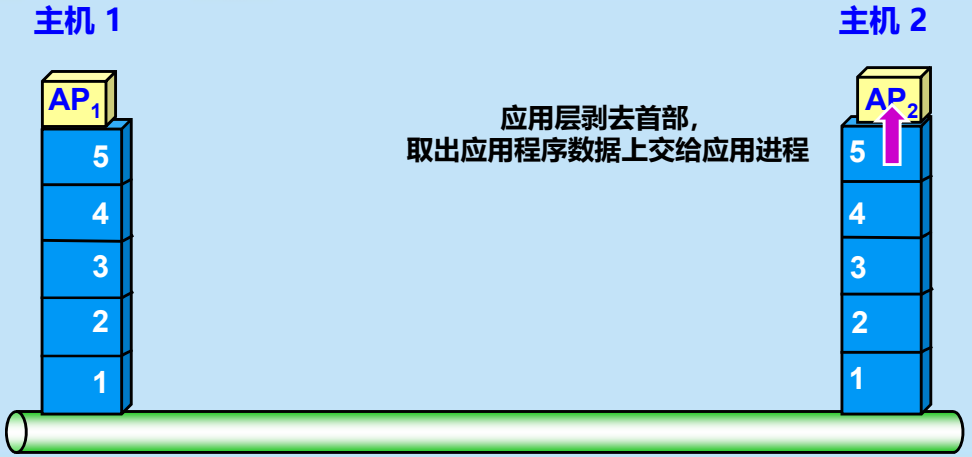
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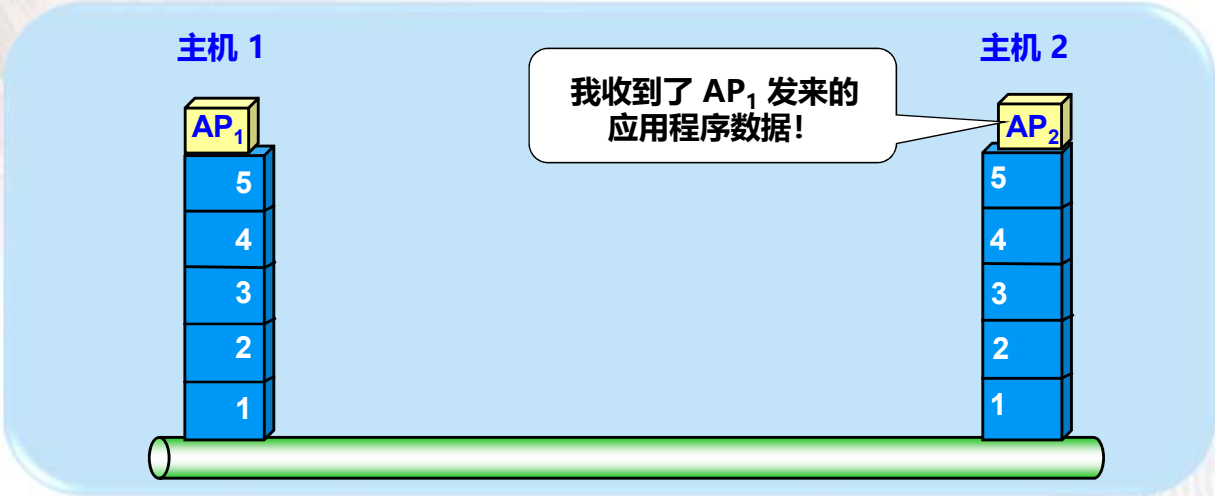
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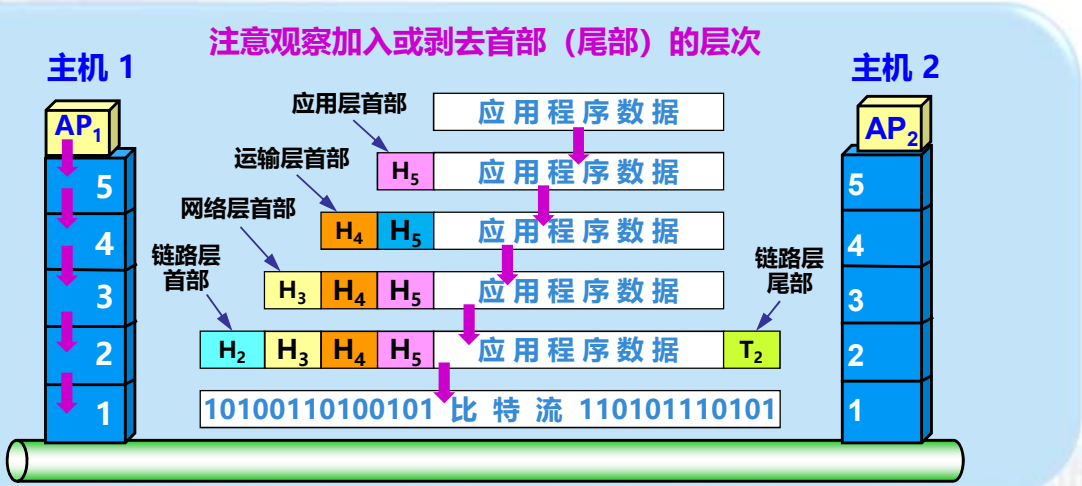
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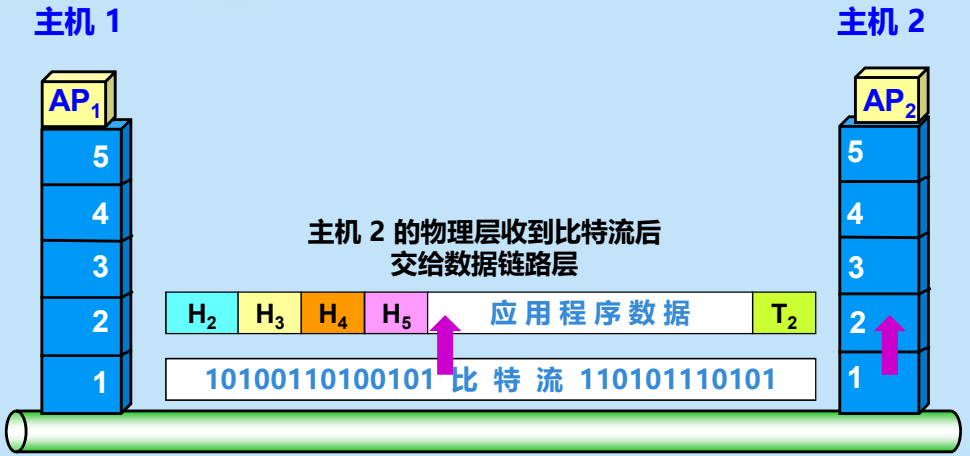
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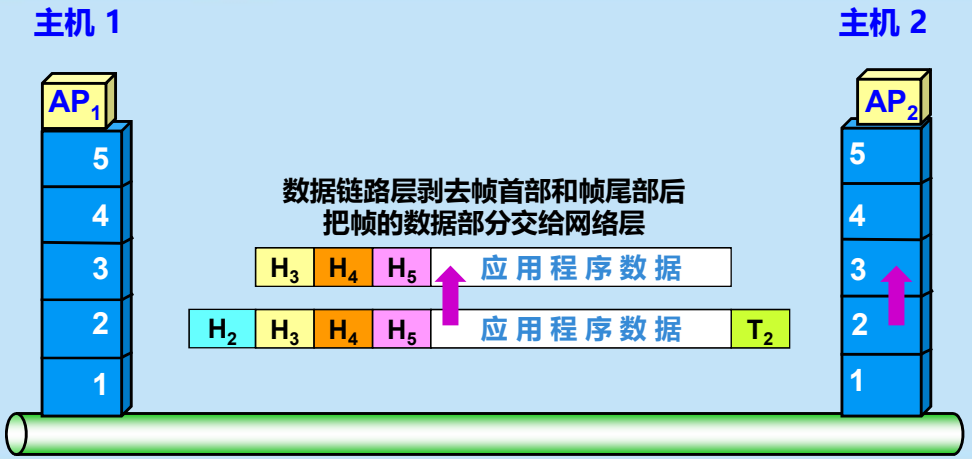
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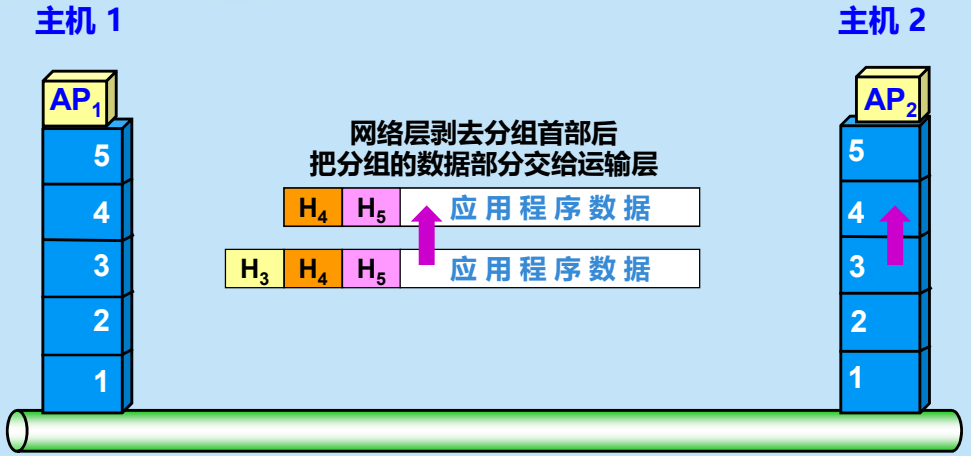
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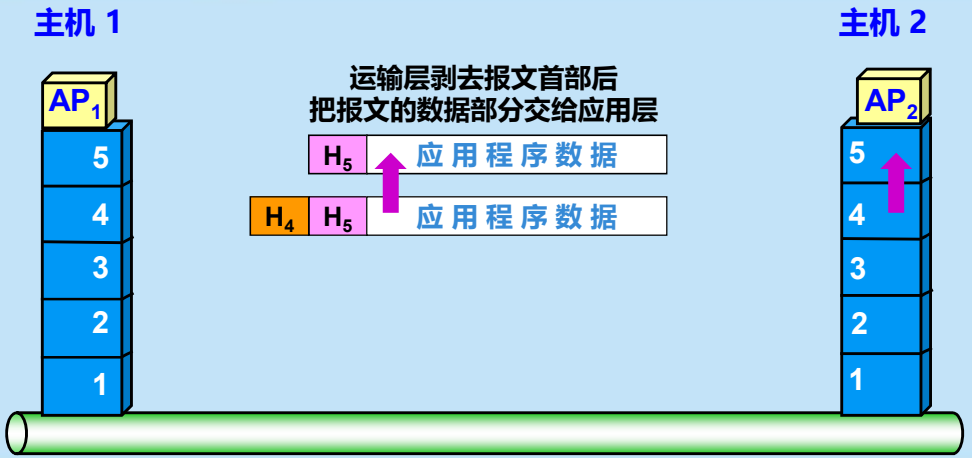
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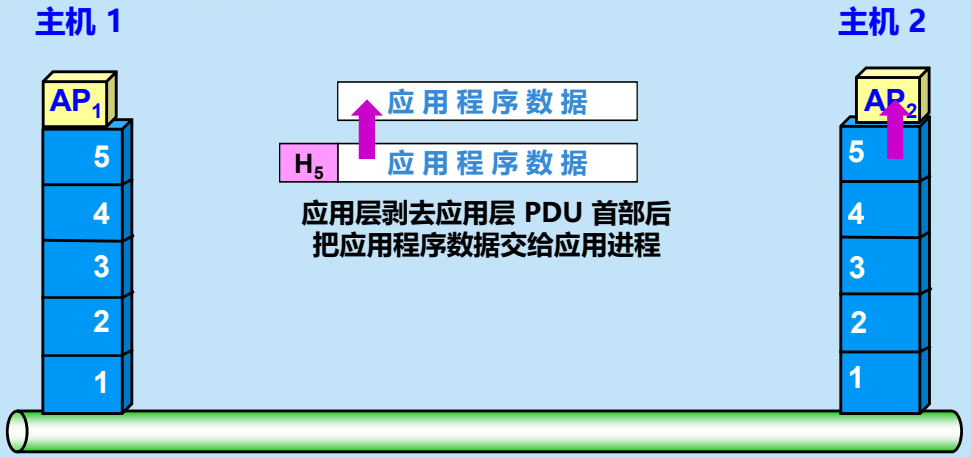
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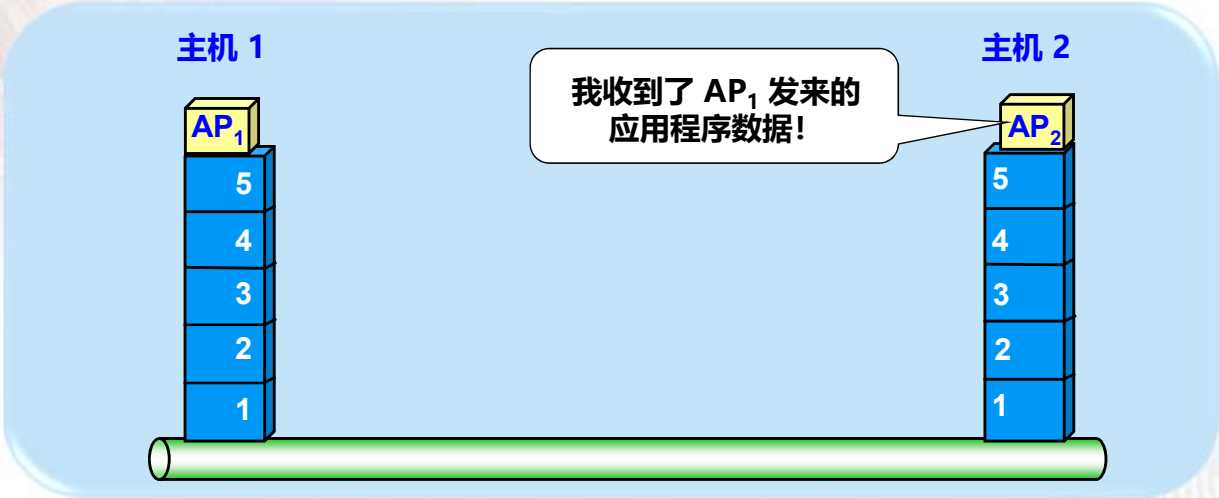
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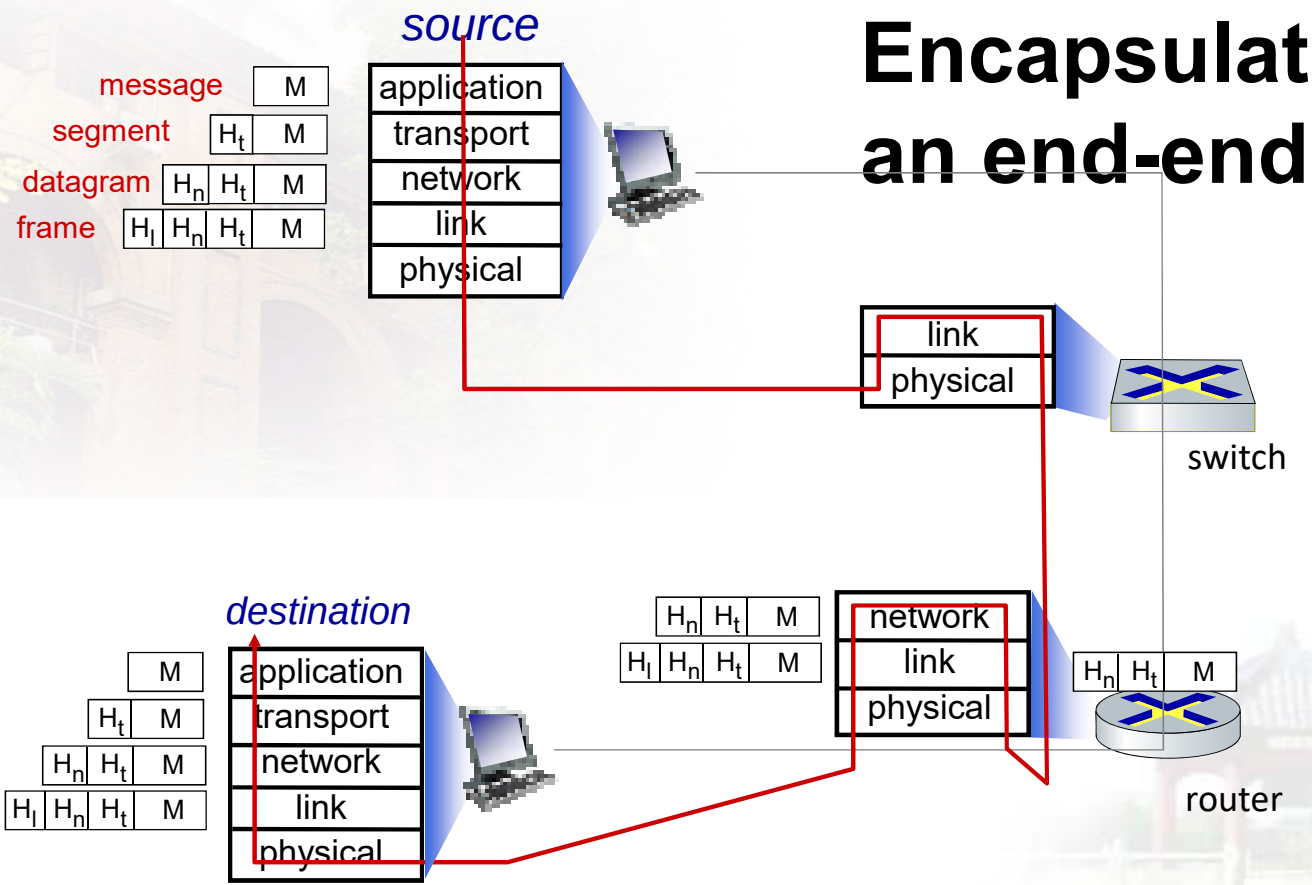
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主机 1 向主机 2 发送数据



Encapsulation: an end-end view



Chapter 1: introduction

1.1 what *is* the Internet?

1.2 network edge

- end systems, access networks, links

1.3 network core

- packet switching, circuit switching, network structure

1.4 delay, loss, throughput in networks

1.5 protocol layers, service models

1.6 networks under attack: security

1.7 history

Network security

- **field of network security:**
  - how bad guys can **attack** computer networks
  - how we can **defend** networks against attacks
  - how to **design** architectures that are immune to attacks
- **Internet not originally designed with (much) security in mind**
  - *original vision:* "a group of mutually trusting users attached to a transparent network"
  - Internet protocol designers playing "catch-up"
  - security considerations in all layers!

Bad guys: put malware into hosts via Internet

- **malware can get in host from:**
  - ***virus:*** **a small program written to alter the way a computer operates, without the permission or knowledge of the user. A virus must meet two criteria:**
    - It must execute itself.
    - It must replicate itself.
  - ***worm:*** **self-replicating infection by passively receiving object that gets itself executed**
  - ***What is the difference between virus and worm?***
    - Virus: Effect host's behavior
    - Worm: travel from computer to computer

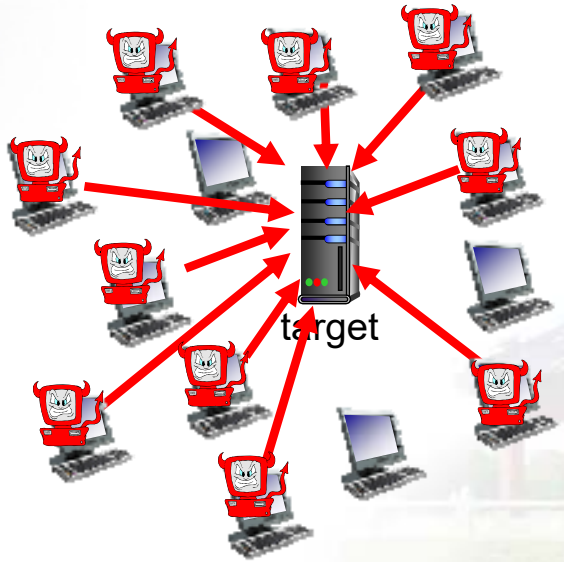
Bad guys: put malware into hosts via Internet

- spyware malware can record keystrokes, web sites visited, upload info to collection site
- infected host can be enrolled in botnet, used for spam. DDoS attacks

Bad guys: attack server, network infrastructure

Denial of Service (DoS): **attackers make resources (server, bandwidth) unavailable to legitimate traffic by overwhelming resource with bogus traffic**

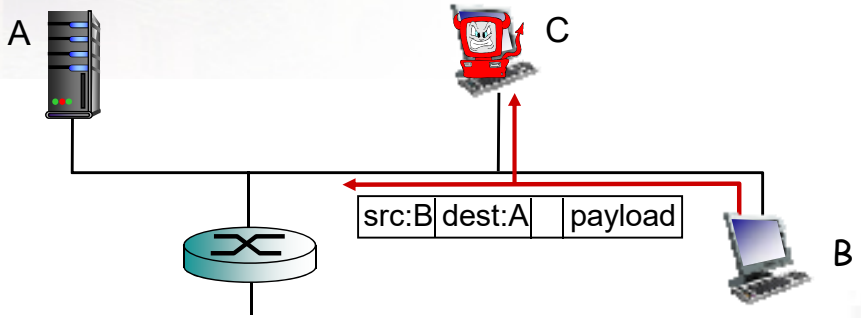
1. select target
2. break into hosts around the network (see botnet)
3. send packets to target from compromised hosts



Bad guys can sniff packets

packet "sniffing":

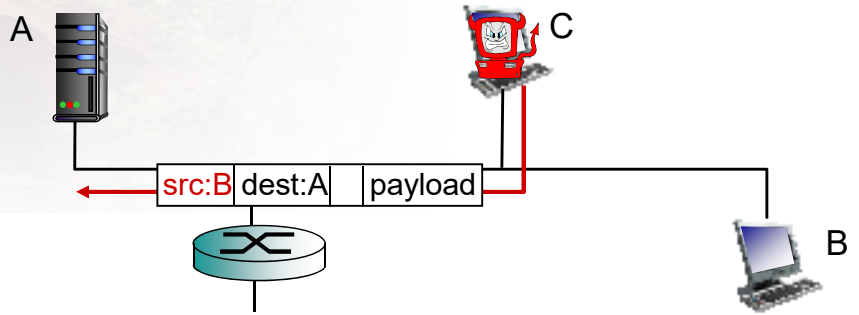
- broadcast media (shared ethernet, wireless)
- promiscuous network interface reads/records all packets (e.g., including passwords!) passing by



- wireshark software used for end-of-chapter labs is a (free) packet-sniffer

Bad guys can use fake addresses

IP spoofing: **send packet with false source address**



… lots more on security (throughout, Chapter 8)

Lines of defense:

- authentication: proving you are who you say you are
  - cellular networks provides hardware identity via SIM card; no such hardware assist in traditional Internet
- confidentiality: via encryption
- integrity checks: digital signatures prevent/detect tampering
- access restrictions: password-protected VPNs
- firewalls: specialized "middleboxes" in access and core networks:
  - off-by-default: filter incoming packets to restrict senders, receivers, applications
  - detecting/reacting to DOS attacks

… lots more on security (throughout, Chapter 8)

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1.5 protocol layers, service models

1.6 networks under attack: security

1.7 history (read after class)

Introduction: summary

covered a "ton" of material!

- Internet overview
- what's a protocol?
- network edge, core, access network
  - packet-switching versus circuit-switching
  - Internet structure
- performance: loss, delay, throughput
- layering, service models
- security
- history

you now have:

- context, overview, "feel" of networking
- more depth, detail *to follow!*

The End of Chapter 1

Thanks

Q & A

Email: xieyi5@mail.sysu.edu.cn
https://cse.sysu.edu.cn/content/2462